

# Corvus Keyword Reference

Automatically Generated

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# 1 Introduction

Corvus is a scientific workflow framework built with python with a focus on calculations of spectroscopy. It is written to facilitate advanced calculations for novice users.

Corvus is "property driven" which means that the user specifies a property to calculate, and Corvus figures out how it can obtain that property, and what inputs are needed from the user. This is done by building a dependency tree, with the root given by the list of requested properties. The dependency tree is built by checking the dependencies of (other properties needed to calculate) the requested property, then asking if these can be satisfied by any of the available codes. This procedure is repeated until either all dependency requirements are met, in which case a workflow is built and run. If all requirements are not met, the user is notified of any necessary user input.

Most of the workflows available to standard users are meant to facilitate calculations using the FEFF10 code. These workflows include loops over specified parameters, configurational averaging, RIXS, and creation of input files from multiple available structural input such as .cif, .xyz, materials project id, or vasp output.

## 2 Download and Installation

### 2.1 Getting Corvus

To get corvus, go to <https://github.com/times-software/Corvus/releases/latest> and download the archived .zip or .tgz file.

### 2.2 Installation

To install corvus, you will need python. Versions  $> 3.12$  and  $< 3.14$  are supported. Earlier versions may work as well. We also strongly suggest that you use an environment manager such as conda to install corvus. Our suggestion is to use the free miniforge package, which can be found at <https://conda-forge.org/download/>

Python's venv environment management package can also be used.

#### 2.2.1 Simple Installation using install script

Once you have a version of Corvus from the github website, you can follow the following simple directions to install. Note that in the below "version" should be replaced with the version that you downloaded, e.g., Corvus-3.2.0-01.zip.

**MacOS** Unzip the downloaded file. Open a terminal and navigate to the file. Run the install.sh script.

```
cd Downloads
unzip Corvus-version.zip
cd Corvus-version/
source install.sh
```

**Windows** Unzip the download file. Open a terminal and navigate to the directory. Run the installer.bat script.

```
cd Downloads
unzip Corvus-version.zip
cd Corvus-version\Corvus-version
install.bat
```

### 2.2.2 Step-by-step instructions with conda

In MacOS, just open a terminal, and it will already have conda enabled if you have it installed. In windows, you will need to open a conda terminal, (Miniforge command window if you installed miniforge conda).

1. Download corvus from the github page.
2. unzip the folder.
3. In the terminal, navigate to the folder
4. Create a conda environment for corvus
5. Install corvus with pip.

```
cd Downloads/Corvus-version/
conda create -name your_corvus_env python=">=3.12,<3.14"
conda activate your_corvus_env
pip install .
```

## 3 Running Corvus

Corvus is run from within a terminal (or miniforge dos command window on windows). If you used the install script to setup corvus, there should be an executable script named "Corvus-version.command" or "Corvus-version.bat" on your Desktop. To run corvus, double click this script, navigate to the input file you want to run, and type,

```
run-corvus -i your_input_file
```

If you installed with the step-by-step instructions, you will need to activate your corvus environment first. With conda, this is done by typing,

```
conda activate your_corvus_env
```

## 4 The Corvus input file

A Corvus input file is composed of a sequence of keyword blocks.

Each keyword is followed by a brace-delimited block containing the data associated with that keyword.

Example:

```
title
{
    Example Calculation
}
```

Supported schema types:

- S = String
- I = Integer
- F = Float
- L = Logical
- P = Free-form text

## 5 The minimal input, and most important keywords

The starting point for any Corvus calculation is to request a property or list of properties to calculate. This is done using the keyword `target_list`. From here, corvus will look for available codes that can calculate each target, and figure out what input is needed. There are many possible targets for Corvus, but here I will list a few of the most important ones for calculating spectra using FEFF10, and how to use them. In general, when using FEFF10 an average over unique sites in the structure must be taken in order to get the full spectrum. Thus in order to perform a spectral calculation using an external structure as input (the suggested method) you must define,

```
target_list{ cfavg } # Calculate the average of some property.
cfavg_target{ xanes } # Average the XANES, could also be xes, rixs, exafs, ... or multiple properties
```

To specify the structure, you can use a variety of file formats, e.g., .xyz, .cif, XDATCAR, or you can have corvus download a structure from the Materials Project. Example:

```
cif_input{ your_structure.cif }
```

You will also need to specify which element and edge you want, e.g., for the Cu K-edge,

```
absorbing_atom_type{ Cu }
feff.edge{ K }
```

Thus a minimal input file for the Cu K-edge of CuO would look as follows:

```
target_list{ cfavg }
cfavg_target{ xanes }
cif_input{ CuO.cif }
absorbing_atom_type{ Cu }
feff.edge{ K }
```

There are of course many useful and important options available, but this will give you a calculation that has reasonable defaults. One special target to note is the "loop" target which allows the user to defined loops over the values associate with any keyword or set of keywords. For single line keywords, the structure is relatively simple:

```
target_list{ loop } # Make corvus produce a loop over some parameters.
loop_target{ cfavg } # The target property to calculate for each calculation is an average
cfavg_target{ xanes } # Average the xanes over different unique sites in the structure.
```

```

# Now define the loop:
loop_parameter{
single_line_keyword
values_line1
values_line2
values_line3
...
}
# Optionally, define labels for the loop directories. Must have the same number of entries as loop_parameter
# loop_labels{
# label1
# label2
# label3
# ...
# }

```

There is also a special value for the first line after "loop\_parameter" that loops over input files. This loops over any defined sets of keyword->value combinations, e.g.,

```

\begin{verbatim}
loop_parameter{
file
file1.in
file2.in
file3.in
...
}

```

This will cause corvus to perform calculations with all parameters loaded from the main input file + any parameters existing in fileN.in. Input in fileN.in overrides input in the main input file.

## 6 The corvus GUI (SciGUI)

SciGUI is a graphical user interface that can be used for Corvus. The installers (install.bat, install.sh) will install it automatically, or you can get it from the github page <https://github.com/times-software/SciGUI/releases/latest>. The GUI can be used to control corvus, open input files, plot output, and setup calculation in an easy manner. If you installed with the install scripts, there will be a launcher on your desktop called Corvus-version\_GUI.bat or Corvus-version\_GUI.command (windows, MacOS). Once launched, you should see a main window with three panels. The left panel shows keyword buttons, which can be selected. The right screen shows help for the selected keyword. The top panel has some controls (run and plot) as well as filters for selecting which keywords to show on the left panel.

### 6.1 General Use

The GUI shows keywords on the left panel. When a user selects a keyword, the right panel shows help for that keyword, along with an "enable" checkbox, and fillable fields for value input. If a user wants to fill a keyword, they check the "enable" checkbox, then fill the fields. When the user

is finished filling out all of the keywords needed for their calculation, they click the run button. SciGUI will then write a corvus input file with all of the enabled keywords filled out, then run corvus on the input file. Note that there are many possible keywords, so there are filters to select keyword by subject or code, and also a search bar to search for keywords. Finally, there is a "Quick Start" button to help users get started with specific types of calculations.

## 6.2 The controls and filter panel (top panel)

This panel has the run button, the Quick Start selection, and a variety of ways to filter and search keywords:

- Filter by code Which code is this a specific keyword for, or is it a general keyword.
- Filter by subject Keywords related to a specific property of subject.
- Search bar Search for specific terms within a keyword
- "Only enabled" checkbox Show only the enabled keywords.

## 6.3 The Quick Start Menu

If a user selects one of the options from the "Quick Start" menu, the `target_list` keyword (and maybe some others) will automatically be enabled and filled out. In addition, keywords will be colored according to importance:

- RED: Required
- BLUE: Most useful
- GRAY: Other/Advanced

By default, when the "Quick Start" option is chosen, only required and useful cards are shown. In order to fill in the minimum requirements for this type of calculation, the user then just needs to click on each of the red keywords shown in the keyword panel, and fill in the values for that keyword.

## 6.4 Validation

If a user has enabled a keyword (selecting it on the left panel, then clicking the "enable" checkbox on the right) then they must fill all required fields or uncheck the enabled checkbox before selecting another keyword. Otherwise a pop-up message will warn the user, and the same keyword will be selected again.

# 7 Output directory structure and output files

The output file and directory structure of Corvus is as follows: Calculations will be run in sub-directories of your current directory named `Corvus_description` where "description" describes what is in the directory.

Simple calculation that ask directly for a spectrum calculated by FEFF are calculated in the directory `Corvus1_FEFF`, while asking for an average over configurations (`target_list{ cfavg }`) produces a top-level directory named, `Corvus3_cfavg_spectrum`, where spectrum is the named

spectrum that was requested by the user, e.g., `Corvus3_cfavg_xanes`. Then each of the subdirectories of `Corvus3_cfavg_xanes`, will have directories named by the index of the absorber as it was run, a label connected to which atom is the absorber, and the code (FEFF in this case).

For the special target loop, the top level folders start with `CorvusN_loop_loopp parameter_loopvalue`. For example, if the user requests a loop over the parameter `feff.corehole`, with values `none`, `rpa`, `fsr`, then Corvus produces the directories `Corvus1_loop_feff.corehole_none`, `Corvus1_loop_feff.corehole_rpa`, `Corvus1_loop_feff.corehole_fsr`

Finally, the main output files of Corvus are named `Corvus.target_subtarget.out`. Thus for calculations that request xanes and provide the structure directly in the input file, the main output is in `Corvus.xanes.out`. For configurational averages (`cfavg`), the main output is in `Corvus.cfavg_xanes.out` (for xanes calculations). Finally for loop calculations, the main output is inside the top-level loop directories, and named the same as it would be for a single calculation, e.g. `Corvus.xanes.out` or `Corvus.cfavg_xes.out` etc.

The documentation below lists every recognized keyword, its expected schema, default value, and documentation string.

## 8 Keyword Index by Section

### Corvus Controls

- `multiprocessing_level`
- `multiprocessing_ncpu`
- `scratch`
- `target_list`
- `title`
- `usehandlers`
- `usesaved`
- `write_input_only`

### General Methodology

- `clusterradius`
- `constant_volume`
- `keep_symm`
- `method`
- `molecule_vacuum_margin`
- `mt_overlap`
- `nkpoints`
- `nqpoints`

- pspfiles
- pw\_encut
- scf\_conv
- xc

## Physical Properties

- absorbing\_atom
- absorbing\_atom\_by\_label
- absorbing\_atom\_type
- atomic\_charge
- cell\_angles\_abc
- cell\_scaling\_abc
- cell\_scaling\_iso
- cell\_struc\_opt\_flags
- cell\_struc\_xyz\_red
- cell\_vectors
- charge
- cif\_input
- cluster
- fermi\_shift
- ismetal
- multiplicity
- number\_of\_atoms
- polarization
- species
- spectral\_broadening
- spin\_moment
- supercell\_dimensions
- vasp\_outcar
- vasp\_snapshot

- vasp\_xdatcar
- vasp\_xml
- xyz\_input
- xyz\_snapshot

## **Abinit**

- abinit.ng2qpt
- abinit.ngqpt
- abinit.verbatim

## **DMDW**

- dmdw.ioflag
- dmdw.nlanc
- dmdw.paths
- dmdw.tempgrid

## **NWChem**

- nwchem.basis
- nwchem.charge
- nwchem.iter
- nwchem.mult
- nwchem.qmd.dt\_nucl
- nwchem.qmd.nstep\_nucl
- nwchem.qmd.print\_xyz
- nwchem.qmd.snapstep
- nwchem.qmd.targ\_temp
- nwchem.qmd.thermostat
- nwchem.xas.alpha
- nwchem.xas.nroots
- nwchem.xas.vec
- nwchem.xas.xrayenergywin
- nwchem.xaselem
- nwchem.xc

## FEFF

- debyetemp
- dmdw\_nlanczos
- feff.absolute
- feff.afolp
- feff.atoms
- feff.cfaverage
- feff.cgrid
- feff.chbroad
- feff.chsh
- feff.chwidth
- feff.cif
- feff.compton
- feff.config
- feff.control
- feff.coordinates
- feff.corehole
- feff.corrections
- feff.corval
- feff.criteria
- feff.crpa
- feff.danes
- feff.debye
- feff.density
- feff.dims
- feff.dmdw
- feff.edge
- feff.egapse
- feff.egrid

- feff.ellipticity
- feff.elnes
- feff.eps0
- feff.equivalence
- feff.equivalence\_nmax
- feff.exafs
- feff.exchange
- feff.exelfs
- feff.extpot
- feff.fms
- feff.folp
- feff.fprime
- feff.hole
- feff.hubbard
- feff.interstitial
- feff.ion
- feff.iorder
- feff.jumpm
- feff.kmesh
- feff.lattice
- feff.ldec
- feff.ldos
- feff.lfms1
- feff.lfms2
- feff.ljmax
- feff.magic
- feff.mbconv
- feff.mpse
- feff.multipole

- feff.nleg
- feff.nohole
- feff.nrixs
- feff.nstar
- feff.numdens
- feff.opcons
- feff.overlap
- feff.pcriteria
- feff.pmbse
- feff.polarization
- feff.potentials
- feff.potentials.spin
- feff.preps
- feff.print
- feff.rconv
- feff.real
- feff.reciprocal
- feff.restart
- feff.rgrid
- feff.rhozzp
- feff.rixs
- feff.rlprint
- feff.rmultiplier
- feff.rpath
- feff.rphases
- feff.rsigma
- feff.s02
- feff.scf
- feff.scframp

- feff.screen
- feff.scxc
- feff.self
- feff.setedge
- feff.sfconv
- feff.sfse
- feff.sgroup
- feff.sig2
- feff.sig3
- feff.siggk
- feff.spin
- feff.ss
- feff.strfac
- feff.symmetry
- feff.target
- feff.tdlda
- feff.temperature
- feff.title
- feff.tolscf
- feff.unfreezef
- feff.xanes
- feff.xes
- feff.xmcd
- nuctemp
- opcons.drude
- opcons.usesaved
- run\_dmdw

## ORCA

- `orca.basis.basis`
- `orca.basis.decontract`
- `orca.ci`
- `orca.citype`
- `orca.coords.units`
- `orca.cpcm`
- `orca.expert`
- `orca.geom.optimizationquality`
- `orca.method.allowrhf`
- `orca.method.amass`
- `orca.method.frozencore`
- `orca.method.grid`
- `orca.method.gridx`
- `orca.method.method`
- `orca.method.ri`
- `orca.method.runtime`
- `orca.method.usesymm`
- `orca.mp2`
- `orca.mp2type`
- `orca.output.pdbfile`
- `orca.output.print`
- `orca.output.printlevel`
- `orca.output.xyzfile`
- `orca.rel.method`
- `orca.rel.soctype`
- `orca.scf.autostart`
- `orca.scf.cnvdamp`
- `orca.scf.cnvshift`

- `orca.scf.convergence`
- `orca.scf.convergencestrategy`
- `orca.scf.diis`
- `orca.scf.fracocc`
- `orca.scf.guess`
- `orca.scf.hftyp`
- `orca.scf.jmatrix`
- `orca.scf.kdiis`
- `orca.scf.keepdens`
- `orca.scf.keepints`
- `orca.scf.kmatrix`
- `orca.scf.maxiter`
- `orca.scf.nr`
- `orca.scf.readints`
- `orca.scf.scfmode`
- `orca.scf.smeartemp`
- `orca.scf.soscf`
- `orca.scf.uno`
- `orca.scf.usecheapints`
- `orca.scf.valformat`
- `orca.tole`
- `orca.tolmaxg`
- `orca.tolrmaxd`
- `orca.tolrmsd`
- `orca.tolrmsg`

## SIESTA

- phsf.broadening
- phsf.ekeqp
- phsf.numtimepoints
- siesta.AtomicCoordinatesFormat
- siesta.Beta.Broadening
- siesta.Block.AtomicCoordinatesAndAtomicSpecies
- siesta.Block.ChemicalSpeciesLabel
- siesta.Block.LatticeParameters
- siesta.Block.PAO.Basis
- siesta.Coreresponse.Broadening
- siesta.Diag.DivideAndConquer
- siesta.DM.InitSpin.AF
- siesta.DM.MixingWeight
- siesta.DM.NumberBroyden
- siesta.DM.NumberBroyden
- siesta.DM.NumberPulay
- siesta.DM.Tolerance
- siesta.ElectronicTemperature
- siesta.expert
- siesta.LatticeConstant
- siesta.LongOutput
- siesta.MaxSCFIterations
- siesta.MeshCutoff
- siesta.NumberOfAtoms
- siesta.NumberOfSpecies
- siesta.PAO.BasisSize
- siesta.PAO.BasisType
- siesta.SolutionMethod

- siesta.SpinPolarized
- siesta.TD.CoreExcitedAtom
- siesta.TD.CorePerturbationCharge
- siesta.TD.mxPC
- siesta.TD.NumberOfTimeSteps
- siesta.TD.ShapeOfEfield
- siesta.TD.TimeStep
- siesta.UseSaveData
- siesta.XC.authors
- siesta.XC.functional

## **cif2cell**

- cif2cell.atomicunits
- cif2cell.cartesian
- cif2cell.cif\_input
- cif2cell.outfile
- cif2cell.program
- cif2cell.supercell

## **Loop Workflows**

- loop\_labels
- loop\_parameter
- loop\_target

## **Configurational Averaging**

- cfavg\_choose\_random\_absorbers
- cfavg\_egrid
- cfavg\_max\_configurations
- cfavg\_max\_configurations
- cfavg\_target

## Fitting

- fit.bond
- fit.datafile
- fit.parameters
- fit.target

## OCEAN

- ocean.abpad
- ocean.acell
- ocean.cnbse.broaden
- ocean.cnbse.gmres.elist
- ocean.cnbse.gmres.erange
- ocean.cnbse.gmres.fff
- ocean.cnbse.gmres.gprc
- ocean.cnbse.gmres.nloop
- ocean.cnbse.mode
- ocean.cnbse.niter
- ocean.cnbse.rad
- ocean.cnbse.solver
- ocean.cnbse.spect\_range
- ocean.cnbse.strength
- ocean.cnbse.xmesh
- ocean.core\_offset
- ocean.degauss
- ocean.dft
- ocean.dft\_energy\_range
- ocean.diemac
- ocean.ecut
- ocean.edges
- ocean.fband

- `ocean.k0`
- `ocean.ldau`
- `ocean.metal`
- `ocean.natom`
- `ocean.nbands`
- `ocean.ngkpt`
- `ocean.nkpt`
- `ocean.nspin`
- `ocean.nstep`
- `ocean.ntypat`
- `ocean.occopt`
- `ocean.opf.fill`
- `ocean.opf.hfkgrid`
- `ocean.opf.opts`
- `ocean.opf.program`
- `ocean.para_prefix`
- `ocean.photon.energy`
- `ocean.photon.operator`
- `ocean.photon.polarization`
- `ocean.photon.qhat`
- `ocean.photon_q`
- `ocean.pp_list`
- `ocean.rprim`
- `ocean.scfac`
- `ocean.scratch`
- `ocean.screen.nbands`
- `ocean.screen.nkpt`
- `ocean.screen.shells`
- `ocean.screen_energy_range`

- ocean.ser\_prefix
- ocean.smag
- ocean.smag.ixc
- ocean.spin-orbit
- ocean.toldfe
- ocean.tolwfr
- ocean.typat
- ocean.xred
- ocean.znucl
- ocean.zymb

## Miscellaneous

- ene\_int
- mac\_diel\_const
- magmom\_by\_label
- mp\_apikey
- mp\_id
- spectralFunction\_file
- xanes\_file

## 9 Alphabetical Keyword Index

- abinit.ng2qpt (Abinit)
- abinit.ngqpt (Abinit)
- abinit.verbatim (Abinit)
- absorbing\_atom (Physical Properties)
- absorbing\_atom\_by\_label (Physical Properties)
- absorbing\_atom\_type (Physical Properties)
- atomic\_charge (Physical Properties)
- cell\_angles\_abc (Physical Properties)
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- `cif2cell.cif_input` (cif2cell)
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- `cif_input` (Physical Properties)
- `cluster` (Physical Properties)
- `clusterradius` (General Methodology)
- `constant_volume` (General Methodology)
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- `dmdw.nlanc` (DMDW)
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- `dmdw_nlanczos` (FEFF)
- `ene_int` (Miscellaneous)
- `feff.absolute` (FEFF)
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- feff.atoms (FEFF)
- feff.cfaverage (FEFF)
- feff.cgrid (FEFF)
- feff.chbroad (FEFF)
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- feff.cif (FEFF)
- feff.compton (FEFF)
- feff.config (FEFF)
- feff.control (FEFF)
- feff.coordinates (FEFF)
- feff.corehole (FEFF)
- feff.corrections (FEFF)
- feff.corval (FEFF)
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- feff.danes (FEFF)
- feff.debye (FEFF)
- feff.density (FEFF)
- feff.dims (FEFF)
- feff.dmdw (FEFF)
- feff.edge (FEFF)
- feff.egapse (FEFF)
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- feff.ellipticity (FEFF)
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- feff.eps0 (FEFF)
- feff.equivalence (FEFF)
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- feff.exafs (FEFF)
- feff.exchange (FEFF)
- feff.exelfs (FEFF)
- feff.extpot (FEFF)
- feff.fms (FEFF)
- feff.folp (FEFF)
- feff.fprime (FEFF)
- feff.hole (FEFF)
- feff.hubbard (FEFF)
- feff.interstitial (FEFF)
- feff.ion (FEFF)
- feff.iorder (FEFF)
- feff.jumprm (FEFF)
- feff.kmesh (FEFF)
- feff.lattice (FEFF)
- feff.ldec (FEFF)
- feff.ldos (FEFF)
- feff.lfms1 (FEFF)
- feff.lfms2 (FEFF)
- feff.ljmax (FEFF)
- feff.magic (FEFF)
- feff.mbconv (FEFF)
- feff.mpse (FEFF)
- feff.multipole (FEFF)
- feff.nleg (FEFF)
- feff.nohole (FEFF)
- feff.nrixs (FEFF)
- feff.nstar (FEFF)
- feff.numdens (FEFF)

- feff.opcons (FEFF)
- feff.overlap (FEFF)
- feff.pcriteria (FEFF)
- feff.pmbse (FEFF)
- feff.polarization (FEFF)
- feff.potentials (FEFF)
- feff.potentials.spin (FEFF)
- feff.preps (FEFF)
- feff.print (FEFF)
- feff.rconv (FEFF)
- feff.real (FEFF)
- feff.reciprocal (FEFF)
- feff.restart (FEFF)
- feff.rgrid (FEFF)
- feff.rhozp (FEFF)
- feff.rixs (FEFF)
- feff.rlprint (FEFF)
- feff.rmultiplier (FEFF)
- feff.rpath (FEFF)
- feff.rphases (FEFF)
- feff.rsigma (FEFF)
- feff.s02 (FEFF)
- feff.scf (FEFF)
- feff.scframp (FEFF)
- feff.screen (FEFF)
- feff.scxc (FEFF)
- feff.self (FEFF)
- feff.setedge (FEFF)
- feff.sfconv (FEFF)

- feff.sfse (FEFF)
- feff.sgroup (FEFF)
- feff.sig2 (FEFF)
- feff.sig3 (FEFF)
- feff.siggk (FEFF)
- feff.spin (FEFF)
- feff.ss (FEFF)
- feff.strfac (FEFF)
- feff.symmetry (FEFF)
- feff.target (FEFF)
- feff.tdlda (FEFF)
- feff.temperature (FEFF)
- feff.title (FEFF)
- feff.tolscf (FEFF)
- feff.unfreezef (FEFF)
- feff.xanes (FEFF)
- feff.xes (FEFF)
- feff.xmcd (FEFF)
- fermi\_shift (Physical Properties)
- fit.bond (Fitting)
- fit.datafile (Fitting)
- fit.parameters (Fitting)
- fit.target (Fitting)
- ismetal (Physical Properties)
- keep\_symm (General Methodology)
- loop\_labels (Loop Workflows)
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- loop\_target (Loop Workflows)
- mac\_diel\_const (Miscellaneous)

- `magmom_by_label` (Miscellaneous)
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- `molecule_vacuum_margin` (General Methodology)
- `mp_apikey` (Miscellaneous)
- `mp_id` (Miscellaneous)
- `mt_overlap` (General Methodology)
- `multiplicity` (Physical Properties)
- `multiprocessing_level` (Corvus Controls)
- `multiprocessing_ncpu` (Corvus Controls)
- `nkpoints` (General Methodology)
- `nqpoints` (General Methodology)
- `nuctemp` (FEFF)
- `number_of_atoms` (Physical Properties)
- `nwchem.basis` (NWChem)
- `nwchem.charge` (NWChem)
- `nwchem.iter` (NWChem)
- `nwchem.mult` (NWChem)
- `nwchem.qmd.dt_nucl` (NWChem)
- `nwchem.qmd.nstep_nucl` (NWChem)
- `nwchem.qmd.print_xyz` (NWChem)
- `nwchem.qmd.snapstep` (NWChem)
- `nwchem.qmd.targ_temp` (NWChem)
- `nwchem.qmd.thermostat` (NWChem)
- `nwchem.xas.alpha` (NWChem)
- `nwchem.xas.nroots` (NWChem)
- `nwchem.xas.vec` (NWChem)
- `nwchem.xas.xrayenergywin` (NWChem)
- `nwchem.xaselem` (NWChem)
- `nwchem.xc` (NWChem)

- `ocean.abpad` (OCEAN)
- `ocean.acell` (OCEAN)
- `ocean.cnbse.broaden` (OCEAN)
- `ocean.cnbse.gmres.elist` (OCEAN)
- `ocean.cnbse.gmres.erange` (OCEAN)
- `ocean.cnbse.gmres.fff` (OCEAN)
- `ocean.cnbse.gmres.gprc` (OCEAN)
- `ocean.cnbse.gmres.nloop` (OCEAN)
- `ocean.cnbse.mode` (OCEAN)
- `ocean.cnbse.niter` (OCEAN)
- `ocean.cnbse.rad` (OCEAN)
- `ocean.cnbse.solver` (OCEAN)
- `ocean.cnbse.spect_range` (OCEAN)
- `ocean.cnbse.strength` (OCEAN)
- `ocean.cnbse.xmesh` (OCEAN)
- `ocean.core_offset` (OCEAN)
- `ocean.degauss` (OCEAN)
- `ocean.dft` (OCEAN)
- `ocean.dft_energy_range` (OCEAN)
- `ocean.diemac` (OCEAN)
- `ocean.ecut` (OCEAN)
- `ocean.edges` (OCEAN)
- `ocean.fband` (OCEAN)
- `ocean.k0` (OCEAN)
- `ocean.ldau` (OCEAN)
- `ocean.metal` (OCEAN)
- `ocean.natom` (OCEAN)
- `ocean.nbands` (OCEAN)
- `ocean.ngkpt` (OCEAN)

- `ocean.nkpt` (OCEAN)
- `ocean.nspin` (OCEAN)
- `ocean.nstep` (OCEAN)
- `ocean.ntypat` (OCEAN)
- `ocean.occopt` (OCEAN)
- `ocean.opf.fill` (OCEAN)
- `ocean.opf.hfkgrid` (OCEAN)
- `ocean.opf.opts` (OCEAN)
- `ocean.opf.program` (OCEAN)
- `ocean.para_prefix` (OCEAN)
- `ocean.photon.energy` (OCEAN)
- `ocean.photon.operator` (OCEAN)
- `ocean.photon.polarization` (OCEAN)
- `ocean.photon.qhat` (OCEAN)
- `ocean.photon_q` (OCEAN)
- `ocean.pp_list` (OCEAN)
- `ocean.rprim` (OCEAN)
- `ocean.scfac` (OCEAN)
- `ocean.scratch` (OCEAN)
- `ocean.screen.nbands` (OCEAN)
- `ocean.screen.nkpt` (OCEAN)
- `ocean.screen.shells` (OCEAN)
- `ocean.screen_energy_range` (OCEAN)
- `ocean.ser_prefix` (OCEAN)
- `ocean.smag` (OCEAN)
- `ocean.smag.ixc` (OCEAN)
- `ocean.spin-orbit` (OCEAN)
- `ocean.toldfe` (OCEAN)
- `ocean.tolwfr` (OCEAN)

- `ocean.typat` (OCEAN)
- `ocean.xred` (OCEAN)
- `ocean.znucl` (OCEAN)
- `ocean.zymb` (OCEAN)
- `opcons.drude` (FEFF)
- `opcons.usesaved` (FEFF)
- `orca.basis.basis` (ORCA)
- `orca.basis.decontract` (ORCA)
- `orca.ci` (ORCA)
- `orca.citype` (ORCA)
- `orca.coords.units` (ORCA)
- `orca.cpcm` (ORCA)
- `orca.expert` (ORCA)
- `orca.geom.optimizationquality` (ORCA)
- `orca.method.allowrhf` (ORCA)
- `orca.method.amass` (ORCA)
- `orca.method.frozencore` (ORCA)
- `orca.method.grid` (ORCA)
- `orca.method.gridx` (ORCA)
- `orca.method.method` (ORCA)
- `orca.method.ri` (ORCA)
- `orca.method.runttype` (ORCA)
- `orca.method.usesymm` (ORCA)
- `orca.mp2` (ORCA)
- `orca.mp2type` (ORCA)
- `orca.output.pdbfile` (ORCA)
- `orca.output.print` (ORCA)
- `orca.output.printlevel` (ORCA)
- `orca.output.xyzfile` (ORCA)

- `orca.rel.method` (ORCA)
- `orca.rel.soctype` (ORCA)
- `orca.scf.autostart` (ORCA)
- `orca.scf.cnvdamp` (ORCA)
- `orca.scf.cnvshift` (ORCA)
- `orca.scf.convergence` (ORCA)
- `orca.scf.convergencestrategy` (ORCA)
- `orca.scf.diis` (ORCA)
- `orca.scf.fracocc` (ORCA)
- `orca.scf.guess` (ORCA)
- `orca.scf.hftyp` (ORCA)
- `orca.scf.jmatrix` (ORCA)
- `orca.scf.kdiis` (ORCA)
- `orca.scf.keepdens` (ORCA)
- `orca.scf.keepints` (ORCA)
- `orca.scf.kmatrix` (ORCA)
- `orca.scf.maxiter` (ORCA)
- `orca.scf.nr` (ORCA)
- `orca.scf.readints` (ORCA)
- `orca.scf.scfmode` (ORCA)
- `orca.scf.smeartemp` (ORCA)
- `orca.scf.soscf` (ORCA)
- `orca.scf.uno` (ORCA)
- `orca.scf.usecheapints` (ORCA)
- `orca.scf.valformat` (ORCA)
- `orca.tole` (ORCA)
- `orca.tolmaxg` (ORCA)
- `orca.tolrmaxd` (ORCA)
- `orca.tolrmsd` (ORCA)

- `orca.tolrmsg` (ORCA)
- `phsf.broadening` (SIESTA)
- `phsf.ekeqp` (SIESTA)
- `phsf.numtimepoints` (SIESTA)
- `polarization` (Physical Properties)
- `pspfiles` (General Methodology)
- `pw_encut` (General Methodology)
- `run_dmdw` (FEFF)
- `scf_conv` (General Methodology)
- `scratch` (Corvus Controls)
- `siesta.AtomicCoordinatesFormat` (SIESTA)
- `siesta.Beta.Broadening` (SIESTA)
- `siesta.Block.AtomicCoordinatesAndAtomicSpecies` (SIESTA)
- `siesta.Block.ChemicalSpeciesLabel` (SIESTA)
- `siesta.Block.LatticeParameters` (SIESTA)
- `siesta.Block.PAO.Basis` (SIESTA)
- `siesta.Coreresponse.Broadening` (SIESTA)
- `siesta.Diag.DivideAndConquer` (SIESTA)
- `siesta.DM.InitSpin.AF` (SIESTA)
- `siesta.DM.MixingWeight` (SIESTA)
- `siesta.DM.NumberBroyden` (SIESTA)
- `siesta.DM.NumberBroyden` (SIESTA)
- `siesta.DM.NumberPulay` (SIESTA)
- `siesta.DM.Tolerance` (SIESTA)
- `siesta.ElectronicTemperature` (SIESTA)
- `siesta.expert` (SIESTA)
- `siesta.LatticeConstant` (SIESTA)
- `siesta.LongOutput` (SIESTA)
- `siesta.MaxSCFIterations` (SIESTA)

- `siesta.MeshCutoff` (SIESTA)
- `siesta.NumberOfAtoms` (SIESTA)
- `siesta.NumberOfSpecies` (SIESTA)
- `siesta.PAO.BasisSize` (SIESTA)
- `siesta.PAO.BasisType` (SIESTA)
- `siesta.SolutionMethod` (SIESTA)
- `siesta.SpinPolarized` (SIESTA)
- `siesta.TD.CoreExcitedAtom` (SIESTA)
- `siesta.TD.CorePerturbationCharge` (SIESTA)
- `siesta.TD.mxPC` (SIESTA)
- `siesta.TD.NumberOfTimeSteps` (SIESTA)
- `siesta.TD.ShapeOfEfield` (SIESTA)
- `siesta.TD.TimeStep` (SIESTA)
- `siesta.UseSaveData` (SIESTA)
- `siesta.XC.authors` (SIESTA)
- `siesta.XC.functional` (SIESTA)
- `species` (Physical Properties)
- `spectral_broadening` (Physical Properties)
- `spectralFunction_file` (Miscellaneous)
- `spin_moment` (Physical Properties)
- `supercell_dimensions` (Physical Properties)
- `target_list` (Corvus Controls)
- `title` (Corvus Controls)
- `usehandlers` (Corvus Controls)
- `usesaved` (Corvus Controls)
- `vasp_outcar` (Physical Properties)
- `vasp_snapshot` (Physical Properties)
- `vasp_xdatcar` (Physical Properties)
- `vasp_xml` (Physical Properties)

- write\_input\_only (Corvus Controls)
- xanes\_file (Miscellaneous)
- xc (General Methodology)
- xyz\_input (Physical Properties)
- xyz\_snapshot (Physical Properties)

## 10 Corvus Controls

### 10.1 multiprocessing\_level

|            |                       |
|------------|-----------------------|
| Keyword    | multiprocessing_level |
| Kind       | scalar                |
| Type       | String                |
| Default    | cfavg                 |
| Deprecated | No                    |

Which level of the calculation do you want to use multiprocessing on. Can be "cfavg" or "loop".

#### Schema

cfavg |X| S

### 10.2 multiprocessing\_ncpu

|            |                      |
|------------|----------------------|
| Keyword    | multiprocessing_ncpu |
| Kind       | scalar               |
| Type       | Integer              |
| Default    | 1                    |
| Deprecated | No                   |

Number of processors to use in multiprocessing. This should be maximum of the number of cpus on a single node.

#### Schema

1 |X| I

### 10.3 scratch

|         |         |
|---------|---------|
| Keyword | scratch |
| Kind    | scalar  |
| Type    | String  |
| Default | .       |

|            |    |
|------------|----|
| Deprecated | No |
|------------|----|

Directory for disk scratch for those Handlers that require large amounts of disk. If the directory is not present or it can not be created, Corvus reverts to the default. NOTE: This input variable is not fully implemented yet.

#### Schema

. |X| S ...

### 10.4 target\_list

|            |                                       |
|------------|---------------------------------------|
| Keyword    | target_list                           |
| Kind       | repeating-record                      |
| Type       | Repeating record with columns: String |
| Default    | Required / No default                 |
| Deprecated | No                                    |

Space separated list with all the target properties requested for this calculation.

#### Schema

|X| S ...

|X| ...

### 10.5 title

|            |                                          |
|------------|------------------------------------------|
| Keyword    | title                                    |
| Kind       | repeating-record                         |
| Type       | Repeating record with columns: Free-form |
| Default    | Required / No default                    |
| Deprecated | No                                       |

Title of this calculation.

#### Schema

This is a Corvus calculation |X| P  
|X| ...

### 10.6 usehandlers

|         |             |
|---------|-------------|
| Keyword | usehandlers |
| Kind    | scalar      |
| Type    | String      |

|            |                       |
|------------|-----------------------|
| Default    | Required / No default |
| Deprecated | No                    |

Explicitly declare what Handlers are to be used in the generation of the Workflow. This helps the current simple Workflow generator when a given target is provided by more than one Handler. Current possible values are: Feff, FeffRixs, Dmdw, Abinit, Vasp, Nwchem, Orca

## Schema

|X| S ...

## 10.7 usesaved

|            |          |
|------------|----------|
| Keyword    | usesaved |
| Kind       | scalar   |
| Type       | Logical  |
| Default    | False    |
| Deprecated | No       |

Use previously calculated data rather than recalculating

## Schema

False |X| L

## 10.8 write\_input\_only

|            |                  |
|------------|------------------|
| Keyword    | write_input_only |
| Kind       | scalar           |
| Type       | Logical          |
| Default    | False            |
| Deprecated | No               |

If True, write input for each of the top level calculations, then exit. Not implemented for all calculations.

## Schema

False |X| L

# 11 General Methodology

## 11.1 clusterradius

|         |               |
|---------|---------------|
| Keyword | clusterradius |
|---------|---------------|

|            |        |
|------------|--------|
| Kind       | scalar |
| Type       | Float  |
| Default    | 12.0   |
| Deprecated | No     |

Radius of clusters created from cif files.

## Schema

12.0 |X| F

### 11.2 constant\_volume

|            |                 |
|------------|-----------------|
| Keyword    | constant_volume |
| Kind       | scalar          |
| Type       | Logical         |
| Default    | True            |
| Deprecated | No              |

Keep the simulation cell volume constant in cell optimizations.

## Schema

True |X| L

### 11.3 keep\_symm

|            |           |
|------------|-----------|
| Keyword    | keep_symm |
| Kind       | scalar    |
| Type       | Logical   |
| Default    | True      |
| Deprecated | No        |

Toggle the preservation of initial symmetry in optimizations.

## Schema

True |X| L

### 11.4 method

|         |        |
|---------|--------|
| Keyword | method |
| Kind    | scalar |
| Type    | String |
| Default | dft    |

|            |    |
|------------|----|
| Deprecated | No |
|------------|----|

Method used to compute the target. This variable is somewhat context-dependent and is currently not fully implemented. Current possible values are: dft, mp2, ccscd

#### Schema

dft |X| S

### 11.5 molecule\_vacuum\_margin

|            |                        |
|------------|------------------------|
| Keyword    | molecule_vacuum_margin |
| Kind       | scalar                 |
| Type       | Float                  |
| Default    | Required / No default  |
| Deprecated | No                     |

Added to the largest distance between two atoms in the molecule before making a unit cell

#### Schema

|X| F

### 11.6 mt\_overlap

|            |                                              |
|------------|----------------------------------------------|
| Keyword    | mt_overlap                                   |
| Kind       | repeating-record                             |
| Type       | Repeating record with columns: String, Float |
| Default    | Required / No default                        |
| Deprecated | No                                           |

set the muffin-tin overlap for different elements.

#### Schema

H 0.8 |X| S F  
|X| ...

### 11.7 nkpoints

|            |                                   |
|------------|-----------------------------------|
| Keyword    | nkpoints                          |
| Kind       | vector                            |
| Type       | Vector: Integer, Integer, Integer |
| Default    | 1 1 1                             |
| Deprecated | No                                |

Define the number of k-point in each direction of the grid for reciprocal space simulations.

## Schema

1 1 1 |X| I I I

### 11.8 nqpoints

|            |                                   |
|------------|-----------------------------------|
| Keyword    | nqpoints                          |
| Kind       | vector                            |
| Type       | Vector: Integer, Integer, Integer |
| Default    | Required / No default             |
| Deprecated | No                                |

Define the number of q-point perturbations in each direction of the grid for density functional perturbation theory (DFPT) simulations.

## Schema

|X| I I I

### 11.9 pspfiles

|            |                                               |
|------------|-----------------------------------------------|
| Keyword    | pspfiles                                      |
| Kind       | repeating-record                              |
| Type       | Repeating record with columns: String, String |
| Default    | Required / No default                         |
| Deprecated | No                                            |

List of pseudopotentials for each atom in the system. Each line contains the label for the atoms (usually the element with the pseudopotential). The required format for the files will depend on the Handler used.

## Schema

|X| S S  
|X| ...

### 11.10 pw\_encut

|            |          |
|------------|----------|
| Keyword    | pw_encut |
| Kind       | scalar   |
| Type       | Float    |
| Default    | 15.0     |
| Deprecated | No       |

Planwave energy cutoff for reciprocal space simulations, in au.

## Schema

15.0 |X| F

### 11.11 scf\_conv

|            |          |
|------------|----------|
| Keyword    | scf_conv |
| Kind       | scalar   |
| Type       | Float    |
| Default    | 1.0e-5   |
| Deprecated | No       |

Convergence threshold for SCF cycles (HF, DFT, etc) in au. The value set by this variable might be internally overridden if the target requires tighter convergence settings.

## Schema

1.0e-5|X| F

### 11.12 xc

|            |        |
|------------|--------|
| Keyword    | xc     |
| Kind       | scalar |
| Type       | String |
| Default    | lda    |
| Deprecated | No     |

Exchange-correlation functional to use if the "method" selected is "dft". Current possible values are: lda, pbe, b3lyp

## Schema

lda |X| S

## 12 Physical Properties

### 12.1 absorbing\_atom

|            |                       |
|------------|-----------------------|
| Keyword    | absorbing_atom        |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

Index of the absorbing atom for core spectroscopies such as XANES, EXAFS and XES. The index counts from one for the first atom in the associated cluster (see above).

## Schema

|X| I ...

### 12.2 absorbing\_atom\_by\_label

|            |                         |
|------------|-------------------------|
| Keyword    | absorbing_atom_by_label |
| Kind       | scalar                  |
| Type       | String                  |
| Default    | Required / No default   |
| Deprecated | No                      |

Set the absorber by the label associated with it in the cif file. Will set all atoms that have a label that starts with this string as absorbers.

## Schema

|X| S ...

### 12.3 absorbing\_atom\_type

|            |                       |
|------------|-----------------------|
| Keyword    | absorbing_atom_type   |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Symbol of atom you would like to calculate the XANES for. Will loop over symmetry unique sites defined in the CIF file denoted by cif\_input (see below).

## Schema

|X| S ...

### 12.4 atomic\_charge

|            |                                              |
|------------|----------------------------------------------|
| Keyword    | atomic_charge                                |
| Kind       | repeating-record                             |
| Type       | Repeating record with columns: String, Float |
| Default    | Required / No default                        |
| Deprecated | No                                           |

charge to be placed on each atoms of a given species. atomic\_symbol charge ...

## Schema

|X| S F

|X| ...

## 12.5 cell\_angles\_abc

|            |                             |
|------------|-----------------------------|
| Keyword    | cell_angles_abc             |
| Kind       | vector                      |
| Type       | Vector: Float, Float, Float |
| Default    | Required / No default       |
| Deprecated | No                          |

Scaling of the a, b and c axes of the simulation cell, in Angstroms.

### Schema

|X| F F F

## 12.6 cell\_scaling\_abc

|            |                             |
|------------|-----------------------------|
| Keyword    | cell_scaling_abc            |
| Kind       | vector                      |
| Type       | Vector: Float, Float, Float |
| Default    | Required / No default       |
| Deprecated | No                          |

Scaling of the a, b and c axes of the simulation cell, in Angstroms.

### Schema

|X| F F F

## 12.7 cell\_scaling\_iso

|            |                  |
|------------|------------------|
| Keyword    | cell_scaling_iso |
| Kind       | scalar           |
| Type       | Float            |
| Default    | 1.0              |
| Deprecated | No               |

Unitless isotropic scaling of the unit cell.

### Schema

1.0 |X| F

## 12.8 cell\_struc\_opt\_flags

|            |                                                          |
|------------|----------------------------------------------------------|
| Keyword    | cell_struc_opt_flags                                     |
| Kind       | repeating-record                                         |
| Type       | Repeating record with columns: Logical, Logical, Logical |
| Default    | Required / No default                                    |
| Deprecated | No                                                       |

Toggle the optimization of the x, y and z coordinates of each atom in the system.

### Schema

```
|X| L L L  
|X| ...
```

## 12.9 cell\_struc\_xyz\_red

|            |                                                            |
|------------|------------------------------------------------------------|
| Keyword    | cell_struc_xyz_red                                         |
| Kind       | repeating-record                                           |
| Type       | Repeating record with columns: String, Float, Float, Float |
| Default    | Required / No default                                      |
| Deprecated | No                                                         |

Structure of the simulation cell in reduced coordinates if the system is extended and has periodic boundary conditions.

### Schema

```
|X| S F F F  
|X| ...
```

## 12.10 cell\_vectors

|            |                               |
|------------|-------------------------------|
| Keyword    | cell_vectors                  |
| Kind       | matrix                        |
| Type       | 3 rows of Float, Float, Float |
| Default    | Required / No default         |
| Deprecated | No                            |

Normalized simulation cell vector directions. These vectors are scaled using the "cell\_scaling\_iso" and "cell\_scaling\_abc" input variables to generate the simulation cell.

### Schema

```
|X| F F F  
|X| F F F  
|X| F F F
```

## 12.11 charge

|            |                       |
|------------|-----------------------|
| Keyword    | charge                |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

Net charge of the system.

### Schema

|X| I

## 12.12 cif\_input

|            |                       |
|------------|-----------------------|
| Keyword    | cif_input             |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

cif input file name.

### Schema

|X| S

## 12.13 cluster

|            |                                                            |
|------------|------------------------------------------------------------|
| Keyword    | cluster                                                    |
| Kind       | repeating-record                                           |
| Type       | Repeating record with columns: String, Float, Float, Float |
| Default    | Required / No default                                      |
| Deprecated | No                                                         |

Structure of the system is the system is not extended (i. e. is a molecule or cluster). The coordinates are in Angstroms, and have xyz format, i.e., Atomic\_symbol1 x y z Atomic\_symbol2 x y z . . .

### Schema

|X| S F F F  
|X| . . .

### 12.14 fermi\_shift

|            |                       |
|------------|-----------------------|
| Keyword    | fermi_shift           |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

Shift of the fermi-energy in eV.

#### Schema

|X| F

### 12.15 ismetal

|            |         |
|------------|---------|
| Keyword    | ismetal |
| Kind       | scalar  |
| Type       | Logical |
| Default    | False   |
| Deprecated | No      |

Toggle whether the system should be treated as a metal or as an insulator.

#### Schema

False |X| L

### 12.16 multiplicity

|            |              |
|------------|--------------|
| Keyword    | multiplicity |
| Kind       | scalar       |
| Type       | Integer      |
| Default    | 0            |
| Deprecated | No           |

Multiplicity of the system.

#### Schema

0 |X| I

### 12.17 number\_of\_atoms

|            |                       |
|------------|-----------------------|
| Keyword    | number_of_atoms       |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

Number of atoms in the system

#### Schema

|X| I

### 12.18 polarization

|            |                                                    |
|------------|----------------------------------------------------|
| Keyword    | polarization                                       |
| Kind       | repeating-record                                   |
| Type       | Repeating record with columns: Float, Float, Float |
| Default    | Required / No default                              |
| Deprecated | No                                                 |

Set any number of different polarization directions to calculate in cartesian coordinates. Example: set to x, y, z, i.e., polarization 1.0 0.0 0.0 0.0 1.0 0.0 0.0 0.0 1.0

If set to 0.0 0.0 0.0 (default), only polarization average will be calculated.

#### Schema

0.0 0.0 0.0 |X| F F F  
|X| ...

### 12.19 species

|            |                         |
|------------|-------------------------|
| Keyword    | species                 |
| Kind       | vector                  |
| Type       | Vector: String, Integer |
| Default    | Required / No default   |
| Deprecated | No                      |

chemical species and number of each species

#### Schema

|X| S I

## 12.20 spectral\_broadening

|            |                     |
|------------|---------------------|
| Keyword    | spectral_broadening |
| Kind       | scalar              |
| Type       | Float               |
| Default    | 0.0                 |
| Deprecated | No                  |

Broadening used in some spectroscopic methods.

### Schema

0.0 |X| F

## 12.21 spin\_moment

|            |                                               |
|------------|-----------------------------------------------|
| Keyword    | spin_moment                                   |
| Kind       | repeating-record                              |
| Type       | Repeating record with columns: Integer, Float |
| Default    | Required / No default                         |
| Deprecated | No                                            |

spin moments of atomic types set spin moment for ferromagnetic systems. Only allows one moment per chemical element. Z spin\_moment

### Schema

|X| I F  
|X| ...

## 12.22 supercell\_dimensions

|            |                                   |
|------------|-----------------------------------|
| Keyword    | supercell_dimensions              |
| Kind       | vector                            |
| Type       | Vector: Integer, Integer, Integer |
| Default    | Required / No default             |
| Deprecated | No                                |

supercell dimensions (number of cells in each direction).

### Schema

|X| I I I

## 12.23 vasp\_outcar

|            |                       |
|------------|-----------------------|
| Keyword    | vasp_outcar           |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

specify outcar file to use as input to corvus.

#### Schema

|X| S

### 12.24 vasp\_snapshot

|            |               |
|------------|---------------|
| Keyword    | vasp_snapshot |
| Kind       | scalar        |
| Type       | Integer       |
| Default    | -1            |
| Deprecated | No            |

specify index of structure (snapshot if MD run) to use.

#### Schema

-1 |X| I

### 12.25 vasp\_xdatcar

|            |                       |
|------------|-----------------------|
| Keyword    | vasp_xdatcar          |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

specify xdatcar file to use as input to corvus.

#### Schema

|X| S

### 12.26 vasp\_xml

|         |          |
|---------|----------|
| Keyword | vasp_xml |
| Kind    | scalar   |
| Type    | String   |

|            |                       |
|------------|-----------------------|
| Default    | Required / No default |
| Deprecated | No                    |

specify vasp output xml file to use as input to corvus.

## Schema

|X| S

### 12.27 xyz\_input

|            |                       |
|------------|-----------------------|
| Keyword    | xyz_input             |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

xyz input file name

## Schema

|X| S

### 12.28 xyz\_snapshot

|            |              |
|------------|--------------|
| Keyword    | xyz_snapshot |
| Kind       | scalar       |
| Type       | Integer      |
| Default    | 1            |
| Deprecated | No           |

for multiple structure xyz files, use this snapshot.

## Schema

1 |X| I

## 13 Abinit

### 13.1 abinit.ng2qpt

|         |                                   |
|---------|-----------------------------------|
| Keyword | abinit.ng2qpt                     |
| Kind    | vector                            |
| Type    | Vector: Integer, Integer, Integer |
| Default | Required / No default             |

|            |    |
|------------|----|
| Deprecated | No |
|------------|----|

Equivalent to Abinit's (anaddb) ng2qpt input variable: Defines the number of q-point perturbations in each direction of the finer grid for density functional perturbation theory (DFPT) simulations. Please refer to the Abinit manual for further details.

#### Schema

|X| I I I

### 13.2 abinit.ngqpt

|            |                                   |
|------------|-----------------------------------|
| Keyword    | abinit.ngqpt                      |
| Kind       | vector                            |
| Type       | Vector: Integer, Integer, Integer |
| Default    | Required / No default             |
| Deprecated | No                                |

Equivalent to Abinit's ngqpt input variable: Defines the number of q-point perturbations in each direction of the grid for density functional perturbation theory (DFPT) simulations, and is that equivalent to the general "nqpoints" Corvus input variable. Please refer to the Abinit manual for further details.

NOTE: This code-specific input variable will be replaced by the general "nqpoints" variable in the future.

#### Schema

|X| I I I

### 13.3 abinit.verbatim

|            |                       |
|------------|-----------------------|
| Keyword    | abinit.verbatim       |
| Kind       | scalar                |
| Type       | Free-form             |
| Default    | Required / No default |
| Deprecated | No                    |

The content of this input variable is passed as is (i. e. "verbatim") into the input of all Abinit calculations, and it is meant to help with any Abinit input that is currently not implemented in Corvus. Thus, it should be used carefully to avoid inconsistencies with the automatically generated input. Users should refer to the Abinit manual for information on this extra input.

#### Schema

|X| P

## 14 DMDW

### 14.1 dmdw.ioflag

|            |                       |
|------------|-----------------------|
| Keyword    | dmdw.ioflag           |
| Kind       | scalar                |
| Type       | Free-form             |
| Default    | Required / No default |
| Deprecated | No                    |

Set the amount of output printed by DMDW. Possible values are: 0: Terse, prints out only the desired result ( $\sigma^2$ ,  $u^2$ , etc). 1: Verbose, prints out the pole frequencies and weights, as well as estimates of the Einstein temperatures for each path. Please refer to the DMDW section of the Feff manual for further details.

NOTE: The current format described below is temporary for compatibility with previous versions of Corvus. This will be changed to "Integer" in the future.

#### Schema

|X| P

### 14.2 dmdw.nlanc

|            |                       |
|------------|-----------------------|
| Keyword    | dmdw.nlanc            |
| Kind       | scalar                |
| Type       | Free-form             |
| Default    | Required / No default |
| Deprecated | No                    |

Set the number of Lanczos poles (i. e. iterations) in DMDW. Larger values usually improve convergence of the target quantity. However, the number of poles should not exceed the dimensions of the subspace spanned by the projection of the path into the appropriate eigenmodes of the Hessian. This means that this variable should always be less than  $3*N-6$ , where N is the number of atoms in the system. In practice, a value of 6-8 is sufficient to obtain converged mean square relative displacements (MSRD, or  $\sigma^2$ ) for EXAFS. For crystallographic mean square displacements (MSD, or  $u^2$ ) at least 16 poles are usually needed. Please refer to the DMDW section of the Feff manual for further details.

NOTE: The current format described below is temporary for compatibility with previous versions of Corvus. This will be changed to "Integer" in the future.

#### Schema

|X| P

### 14.3 dmdw.paths

|            |                       |
|------------|-----------------------|
| Keyword    | dmdw.paths            |
| Kind       | scalar                |
| Type       | Free-form             |
| Default    | Required / No default |
| Deprecated | No                    |

Set up the path descriptors that will generate the list of paths for which properties will be calculated. The list has the following form: <Number of descriptors> <Descriptor 1> <Descriptor 2> . .

Each descriptor has the form: <Number of atoms in path> <Atom index 1> ... <Max. path length (Ang)>

The atom indices can take the value 0 which acts as a wildcard for all atoms in the system.

Examples:

2 2 1 0 3.0 3 2 0 5 6.0

This section defines 2 paths descriptors. The first one generates all paths with two atoms, starting in atom 1 and going to all other atoms in the systems, but subject to a maximum effective path length of 3.0 Ang. The second one generates all paths with three atoms, starting in atom 2 and ending in atom 5, while passing through all other atoms in the system, but with a maximum effective length of 6.0 Ang. Using the same syntax atoms can be selected to compute their  $u^2$ . For example the paths section

1 1 0 0.0

will produce  $u^2$  for all atoms in the system. Please refer to the DMDW section of the Feff manual for further details.

NOTE: The current format described below is temporary for compatibility with previous versions of Corvus.

## Schema

|X| P

### 14.4 dmdw.tempgrid

|            |                       |
|------------|-----------------------|
| Keyword    | dmdw.tempgrid         |
| Kind       | scalar                |
| Type       | Free-form             |
| Default    | Required / No default |
| Deprecated | No                    |

Set up the temperature grid to compute the thermal properties in DMDW. It has the following form:

<Number of temperature> <Min. temp.> <Max. temp.>

If the number of desired temperatures is just one, the maximum temperature input is not needed. Please refer to the DMDW section of the Feff manual for further details.

NOTE: The current format described below is temporary for compatibility with previous versions of Corvus.

## Schema

|X| P

# 15 NWChem

## 15.1 nwchem.basis

|            |                                       |
|------------|---------------------------------------|
| Keyword    | nwchem.basis                          |
| Kind       | repeating-record                      |
| Type       | Repeating record with columns: String |
| Default    | Required / No default                 |
| Deprecated | No                                    |

Set the Gaussian basis set to be used in the NWChem calculations. The format is the same as in NWChem:

<Atom label> <Basis set name> . .

Please refer to the NWChem manual for further details.

## Schema

|X| S ...

|X| ...

## 15.2 nwchem.charge

|            |                       |
|------------|-----------------------|
| Keyword    | nwchem.charge         |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Set the net charge of the system for an NWChem DFT simulation. NOTE: This input variable will be superseded in the future by the more general "charge" variable.

Please refer to the NWChem manual for further details.

## Schema

|X| S

## 15.3 nwchem.iter

|         |                       |
|---------|-----------------------|
| Keyword | nwchem.iter           |
| Kind    | scalar                |
| Type    | String                |
| Default | Required / No default |

|            |    |
|------------|----|
| Deprecated | No |
|------------|----|

## Schema

|X| S

### 15.4 nwchem.mult

|            |                       |
|------------|-----------------------|
| Keyword    | nwchem.mult           |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Set the multiplicity of the system for an NWChem DFT simulation. NOTE: This input variable will be superseded in the future by the more general "multiplicity" variable.

Please refer to the NWChem manual for further details.

## Schema

|X| S

### 15.5 nwchem.qmd.dt\_nucl

|            |                       |
|------------|-----------------------|
| Keyword    | nwchem.qmd.dt_nucl    |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Set the time step for nuclear motion in a quantum MD simulation.

Please refer to the NWChem manual for further details.

## Schema

|X| S

### 15.6 nwchem.qmd.nstep\_nucl

|            |                       |
|------------|-----------------------|
| Keyword    | nwchem.qmd.nstep_nucl |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Set the number of nuclear motion steps in a quantum MD simulation.

Please refer to the NWChem manual for further details.

## Schema

|X| S

### 15.7 nwchem.qmd.print\_xyz

|            |                       |
|------------|-----------------------|
| Keyword    | nwchem.qmd.print_xyz  |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Toggle printing of xyz coordinates in a quantum MD simulation.  
Please refer to the NWChem manual for further details.

## Schema

|X| S

### 15.8 nwchem.qmd.snapstep

|            |                       |
|------------|-----------------------|
| Keyword    | nwchem.qmd.snapstep   |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

## Schema

|X| S

### 15.9 nwchem.qmd.targ\_temp

|            |                       |
|------------|-----------------------|
| Keyword    | nwchem.qmd.targ_temp  |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Set the target temperature in a quantum MD simulation.  
Please refer to the NWChem manual for further details.

## Schema

|X| S

### 15.10 nwchem.qmd.thermostat

|            |                       |
|------------|-----------------------|
| Keyword    | nwchem.qmd.thermostat |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Set the thermostat type to be used in a quantum MD simulation.  
Please refer to the NWChem manual for further details.

#### Schema

|X| S

### 15.11 nwchem.xas.alpha

|            |                       |
|------------|-----------------------|
| Keyword    | nwchem.xas.alpha      |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| S

### 15.12 nwchem.xas.nroots

|            |                       |
|------------|-----------------------|
| Keyword    | nwchem.xas.nroots     |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| S

### 15.13 nwchem.xas.vec

|         |                       |
|---------|-----------------------|
| Keyword | nwchem.xas.vec        |
| Kind    | scalar                |
| Type    | String                |
| Default | Required / No default |

|            |    |
|------------|----|
| Deprecated | No |
|------------|----|

#### Schema

|X| S

### 15.14 nwchem.xas.xrayenergywin

|            |                          |
|------------|--------------------------|
| Keyword    | nwchem.xas.xrayenergywin |
| Kind       | vector                   |
| Type       | Vector: Float, Float     |
| Default    | Required / No default    |
| Deprecated | No                       |

#### Schema

|X| F F

### 15.15 nwchem.xaselem

|            |                       |
|------------|-----------------------|
| Keyword    | nwchem.xaselem        |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| S

### 15.16 nwchem.xc

|            |                       |
|------------|-----------------------|
| Keyword    | nwchem.xc             |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Set the exchange correlation potential for an NWChem DFT simulation. NOTE: This input variable will be superseded in the future by the more general "xc" variable.

Please refer to the NWChem manual for further details.

#### Schema

|X| S

## 16 FEFF

### 16.1 debyetemp

|            |                       |
|------------|-----------------------|
| Keyword    | debyetemp             |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

debyetemp temp

Specify the Debye Model temperature (in K) for the calculation of EXAFS DW factors.

#### Schema

|X| F

### 16.2 dmdw\_nlanczos

|            |               |
|------------|---------------|
| Keyword    | dmdw_nlanczos |
| Kind       | scalar        |
| Type       | Integer       |
| Default    | 6             |
| Deprecated | No            |

dmdw\_nlanczos nLanczos\_Interations

Specify the number of Lanczos iterations to be done for the calculation of EXAFS DW factors with the dynamical matrix method.

#### Schema

6 |X| I

### 16.3 feff.absolute

|            |                       |
|------------|-----------------------|
| Keyword    | feff.absolute         |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.absolute UseAbsoluteUnits Print spectrum in absolute units instead of normalizing.

#### Schema

|X| L

## 16.4 feff.afolp

|            |                       |
|------------|-----------------------|
| Keyword    | feff.afolp            |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

feff.afolp folpx Set maximum overlap for automatic overlap search. folpx - maximum overlap allowed.

### Schema

|X| F

## 16.5 feff.atoms

|            |                                                             |
|------------|-------------------------------------------------------------|
| Keyword    | feff.atoms                                                  |
| Kind       | repeating-record                                            |
| Type       | Repeating record with columns: Float, Float, Float, Integer |
| Default    | Required / No default                                       |
| Deprecated | No                                                          |

feff.atoms x1 y1 z1 ipot1 x2 y2 z2 ipot2 . . .

Specify atomic positions in cartesian coordinates (in Angstroms) and unique potential indices of each atom in the cluster, one atom per line.

### Schema

|X| F F F I  
|X| . . .

## 16.6 feff.cfaverage

|            |                                 |
|------------|---------------------------------|
| Keyword    | feff.cfaverage                  |
| Kind       | vector                          |
| Type       | Vector: Integer, Integer, Float |
| Default    | Required / No default           |
| Deprecated | No                              |

Obsolete: Do not use.

### Schema

|X| I I F

## 16.7 feff.cgrid

|            |                                                   |
|------------|---------------------------------------------------|
| Keyword    | feff.cgrid                                        |
| Kind       | vector                                            |
| Type       | Vector: Float, Integer, Integer, Integer, Integer |
| Default    | Required / No default                             |
| Deprecated | No                                                |

Not implemented yet.

### Schema

|X| F I I I I

## 16.8 feff.chbroad

|            |                       |
|------------|-----------------------|
| Keyword    | feff.chbroad          |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.chbroad ichbroad Set method of calculating core-hole lifetime broadening. ichbroad - 0: Calculate Green's function at energy with imaginary part equal to  $\Gamma_{CH}/2$ . 1: Convolve final spectrum with Lorentzian of full width at half-max  $\Gamma_{CH}$ .

### Schema

|X| I

## 16.9 feff.chsh

|            |                       |
|------------|-----------------------|
| Keyword    | feff.chsh             |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.chsh ich Correct chemical shift. ich = 0 or 1.

### Schema

|X| I

## 16.10 feff.chwidth

|            |                       |
|------------|-----------------------|
| Keyword    | feff.chwidth          |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

feff.chwidth gamma\_ch Set core-hole width instead of using FEFF's internal table of widths.

## Schema

|X| F

### 16.11 feff.cif

|            |                       |
|------------|-----------------------|
| Keyword    | feff.cif              |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Specify a cif input file. Dangerous. Use cif\_input instead.

## Schema

|X| S

### 16.12 feff.compton

|            |                                 |
|------------|---------------------------------|
| Keyword    | feff.compton                    |
| Kind       | vector                          |
| Type       | Vector: Float, Integer, Integer |
| Default    | Required / No default           |
| Deprecated | No                              |

Not implemented yet.

## Schema

|X| F I I

### 16.13 feff.config

|         |                                       |
|---------|---------------------------------------|
| Keyword | feff.config                           |
| Kind    | repeating-record                      |
| Type    | Repeating record with columns: String |

|            |                       |
|------------|-----------------------|
| Default    | Required / No default |
| Deprecated | No                    |

Not implemented yet.

## Schema

|X| S  
|X| ...

## 16.14 feff.control

|            |                                                              |
|------------|--------------------------------------------------------------|
| Keyword    | feff.control                                                 |
| Kind       | vector                                                       |
| Type       | Vector: Integer, Integer, Integer, Integer, Integer, Integer |
| Default    | Required / No default                                        |
| Deprecated | No                                                           |

feff.control ipot ixsph ifms ipaths igenfmt iff2x

feff.control lets you run one or more of the feff program modules separately. There is a switch for each of six parts of feff: 0 means not to run that module, 1 means to run it.

## Schema

|X| I I I I I I

## 16.15 feff.coordinates

|            |                       |
|------------|-----------------------|
| Keyword    | feff.coordinates      |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.coordinates i i must be an integer from 1 through 6. It specifies the units of the atoms of the unit cell given in the ATOMS card for reciprocal space calculations. If the card is omitted, the default value icoord = 3 is assumed. FIX check this 1. Cartesian coordinates, Angstrom units. Like feff - you can copy from a real-space feff.inp file if your lattice vectors coincide with atoms in that feff.inp file. 2. Cartesian coordinates, fractional units (i.e., fractions of the lattice vectors ; should be numbers between 0 and 1). Similar to feff. 3. Cartesian coordinates, units are fractional with respect to FIRST lattice vector. Like SPRKKR. (default) 4. Given in lattice coordinates, in fractional units. Like WIEN2k (but beware of some ‘funny’ lattice types, e.g. rhombohedral, in WIEN2k case.struct if you’re copy-pasting ) 5. Given in lattice coordinates, units are fractional with respect to FIRST lattice vector. 6. Given in lattice coordinates, Angstrom units.

## Schema

|X| I

## 16.16 feff.corehole

|            |                       |
|------------|-----------------------|
| Keyword    | feff.corehole         |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

feff.corehole CoreHoleApproximation Set method for calculating corehole interaction. CoreHoleApproximation - None: Do not use a corehole. FSR : Final-state rule RPA : Use RPA dielectric function to calculate screening of core-hole.

### Schema

|X| S

## 16.17 feff.corrections

|            |                       |
|------------|-----------------------|
| Keyword    | feff.corrections      |
| Kind       | vector                |
| Type       | Vector: Float, Float  |
| Default    | Required / No default |
| Deprecated | No                    |

feff.corrections vrcorr vicorr Correct the Fermi cutoff and broadening in the final spectrum calculation. vrcorr - Shift Fermi cutoff by -vrcorr. vicorr - Add extra Lorentzian broadening with half width at half max of vicorr.

### Schema

|X| F F

## 16.18 feff.corval

|            |                       |
|------------|-----------------------|
| Keyword    | feff.corval           |
| Kind       | vector                |
| Type       | Vector: Float, Float  |
| Default    | Required / No default |
| Deprecated | No                    |

feff.corval emin emax The core-valence separation energy is found by scanning the DOS within an energy window. The emin parameter sets the lower bound (in eV) of this energy window and emax the upper. They are -70 eV and -20 eV by default. For some materials it is necessary to change these bounds, e.g. to -100 eV and -30 or -10 eV. For example, when SCF convergence is elusive because occupation numbers for one or more l-values are changing drastically between SCF iterations due to states moving above and below a poor estimate of the core-valence separation energy. We plan to replace the current mechanism by a more robust and automated algorithm, but in the meantime users can use the CORVAL card to handle some of these difficult cases.

## Schema

|X| F F

### 16.19 feff.criteria

|            |                       |
|------------|-----------------------|
| Keyword    | feff.criteria         |
| Kind       | vector                |
| Type       | Vector: Float, Float  |
| Default    | Required / No default |
| Deprecated | No                    |

feff.criteria critcw critpw Cutoff criteria for the path expansion filtering. critcw - tolerance for curved wave expansion critpw - tolerance for initial plane wave approximation

## Schema

|X| F F

### 16.20 feff.crpa

|            |                        |
|------------|------------------------|
| Keyword    | feff.crpa              |
| Kind       | vector                 |
| Type       | Vector: Integer, Float |
| Default    | Required / No default  |
| Deprecated | No                     |

Not currently working.

## Schema

|X| I F

### 16.21 feff.danes

|            |                             |
|------------|-----------------------------|
| Keyword    | feff.danes                  |
| Kind       | vector                      |
| Type       | Vector: Float, Float, Float |
| Default    | Required / No default       |
| Deprecated | No                          |

feff.danes xkmax xkstep estep Calculate diffraction anomalous near edge structure. xkmax - calculate up to  $k = xkmax$  xkstep - use steps of size xkstep estep - near the edge, use steps calculated from estep

## Schema

|X| F F F

## 16.22 feff.debye

|            |                       |
|------------|-----------------------|
| Keyword    | feff.debye            |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

feff.debye Temperature Debye-Temperature idwopt Set temperature and Debye temperature for calculations of EXAFS Debye-Waller factors. idwopt - set method for calculating DW factors. NOTE: This input variable is not fully implemented yet.

### Schema

```
|X| S ...
```

## 16.23 feff.density

|            |                                |
|------------|--------------------------------|
| Keyword    | feff.density                   |
| Kind       | block                          |
| Type       | Multi-line heterogeneous block |
| Default    | Required / No default          |
| Deprecated | No                             |

Not implemented yet.

### Schema

```
# den_type, grid_type, outfile
|X| S S S
# origin
|X| F F F
# number of points in each direction
|X| I I I
# vectors along which to step
|X| F F F
|X| F F F
|X| F F F
```

## 16.24 feff.dims

|            |                          |
|------------|--------------------------|
| Keyword    | feff.dims                |
| Kind       | vector                   |
| Type       | Vector: Integer, Integer |
| Default    | Required / No default    |
| Deprecated | No                       |

feff.dims nmax lmax Set maximum dimensions for fms calculations. nmax - maximum number of atoms in fms matrix. lmax - maximum angular momentum of fms matrix.

## Schema

|X| I I

### 16.25 feff.dmdw

|            |                                           |
|------------|-------------------------------------------|
| Keyword    | feff.dmdw                                 |
| Kind       | vector                                    |
| Type       | Vector: String, Integer, Integer, Integer |
| Default    | Required / No default                     |
| Deprecated | No                                        |

NOTE: This input variable is not fully implemented yet.

## Schema

|X| S I I I

### 16.26 feff.edge

|            |                       |
|------------|-----------------------|
| Keyword    | feff.edge             |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

feff.edge edgeLabel Set edge to calculate for XAS, XES, EELS, and other related spectra.

## Schema

|X| S ...

### 16.27 feff.egapse

|            |                       |
|------------|-----------------------|
| Keyword    | feff.egapse           |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

feff.egapse egap Use this gap energy when applying the many-pole self-energy.

## Schema

|X| F

## 16.28 feff.egrid

|            |                                                             |
|------------|-------------------------------------------------------------|
| Keyword    | feff.egrid                                                  |
| Kind       | repeating-record                                            |
| Type       | Repeating record with columns: String, String, Float, Float |
| Default    | Required / No default                                       |
| Deprecated | No                                                          |

This card can be used to customize the energy grid. The EGRID card is followed by lines specifying the type of grid, minimum and maximum values for the grid, and the grid step, i.e.

grid\_type grid\_min grid\_max grid\_step

The grid type parameter is a string that can take the values e\_grid, k\_grid, or exp\_grid. When using the e\_grid or k\_grid grid types, grid\_min, grid\_max, and grid\_step are given in eV or  $^{-1}$  respectively. For the exp\_grid type, grid\_min and grid\_max are the minimum and maximum grid values in eV, and grid\_step is the exponential, i.e.

$$E_i = \text{grid\_min} + (\text{grid\_max} - \text{grid\_min})[\exp(\text{grid\_step} \times i) - 1].$$

A fourth grid type user grid is also available for feff. user grid is followed by an arbitrary number of lines, each specifying an energy point in eV, i.e., user\_grid 0.1 1.5 3.45 6.0 . . .

### Schema

```
|X| S S F F
|X| ...
```

## 16.29 feff.ellipticity

|            |                                    |
|------------|------------------------------------|
| Keyword    | feff.ellipticity                   |
| Kind       | vector                             |
| Type       | Vector: Float, Float, Float, Float |
| Default    | Required / No default              |
| Deprecated | No                                 |

feff.ellipticity elpty x y z Set the ellipticity of the x-rays. This card is used with the feff.polarization. elpty - ratio of amplitudes of electric field in the two orthogonal directions of elliptically polarized light. Only the absolute value of the ratio is important for nonmagnetic materials. The present code can distinguish left- and right-circular polarization only with the feff.xmcd or feff.xncd. A zero value of the ellipticity corresponds to linear polarization, and unity to circular polarization. The default value is zero. x, y, z - coordinates of any nonzero vector in the direction of the incident beam. This vector should be approximately normal to the polarization vector

### Schema

```
|X| F F F F
```

### 16.30 feff.elnes

|            |                                                    |
|------------|----------------------------------------------------|
| Keyword    | feff.elnes                                         |
| Kind       | repeating-record                                   |
| Type       | Repeating record with columns: Float, Float, Float |
| Default    | Required / No default                              |
| Deprecated | No                                                 |

This card is not implements yet.

#### Schema

```
|X| F F F
|X| S ...
|X| ...
# In future, fix so that we actually have these params, not just strings.
```

### 16.31 feff.eps0

|            |                       |
|------------|-----------------------|
| Keyword    | feff.eps0             |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

feff.eps0 eps0 Set dielectric constant used for many-pole self-energy calculations.

#### Schema

```
|X| F
```

### 16.32 feff.equivalence

|            |                  |
|------------|------------------|
| Keyword    | feff.equivalence |
| Kind       | scalar           |
| Type       | Integer          |
| Default    | 0                |
| Deprecated | No               |

feff.equivalence ieq This optional card is only active in combination with the CIF card. It tells feff how to generate potential types from the list of atom positions in the ‘cif’ file. If ieq = 1, the crystallographic equivalence as expressed in the ‘cif’ file is respected; that is, every separate line containing a generating atom position will lead to a separate potential type. This means that, e.g., in HOPG graphite, the two generating positions will give rise to two independent C potentials. This is also the default behavior if the EQUIVALENCE card is not specified. If ieq = 2, unique potentials are assigned based on atomic number Z only. That is, all C atoms will share a C potential and so on. This is how most feff calculations are run. Whether it is sensible or not to do this depends on

the system and on the property one wishes to calculate. Keep in mind that feff is a muffin tin code, and may therefore be indifferent to certain differences between crystallographically inequivalent sites. On the other hand, if an element occurs in the crystal with different oxidation states, it may be necessary to assign separate potentials to these different types in order to describe the crystal properly and get accurate spectra. If  $ieq = 3$ , unique potentials are assigned based on atomic number  $Z$  and the first shell. This can be useful e.g. to treat larger systems with crystal defects, where only first neighbors of the defect need to be treated differently from all more distant atoms of a certain  $Z$ . (To be implemented.) If  $ieq = 4$ , a hybrid of methods 1 and 2 is used. That is, if the number of unique crystallo- graphic positions does not exceed a hard-coded limit ( $nphx=9$  in the current version), they are treated with the correct crystallographic equivalence. If the number of unique crystallographi- cally inequivalent sites is larger, they get combined by atomic number  $Z$ . This ad hoc approach is a practical way of simply limiting the number of unique potentials. This makes sense be- cause, first of all, there are certain hardcoded limits that would require recompilation of the code, requiring more RAM memory and more work than a user may want to do.

#### Schema

0 |X| I

### 16.33 feff.equivalence\_nmax

|            |                       |
|------------|-----------------------|
| Keyword    | feff.equivalence_nmax |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| I

### 16.34 feff.exafs

|            |                       |
|------------|-----------------------|
| Keyword    | feff.exafs            |
| Kind       | vector                |
| Type       | Vector: Float, Float  |
| Default    | Required / No default |
| Deprecated | No                    |

feff.exafs  $xkmax$   $xkstep$  Calculate EXAFS spectrum.  $xkmax$  - calculate spectrum up to  $k = xkmax$   $xkstep$  - calculate in steps of  $xkstep$

#### Schema

|X| F F

### 16.35 feff.exchange

|            |                                        |
|------------|----------------------------------------|
| Keyword    | feff.exchange                          |
| Kind       | vector                                 |
| Type       | Vector: Integer, Float, Float, Integer |
| Default    | Required / No default                  |
| Deprecated | No                                     |

feff.exchange iexc vr vi iexc0 Use feff.exchange to change the self-energy used in x-ray absorption calculations. ixc is an index specifying the potential model to use for the fine structure, and the optional ixc0 is the index of the model to use for the background function. The calculated potential can be corrected by adding a constant shift to the Fermi level given by vr0 and to a pure imaginary "optical" potential (i.e., uniform decay) given by vi0. Typical errors in Feff's self-consistent Fermi level estimate are about 1 eV. (The feff.corrections input is similar but allows the user to make small changes in vi0 and vr0 after the rest of the calculation is completed, for example in a fitting process.)

Indices for the available exchange models: 0 Hedin-Lundqvist + a constant imaginary part 1 Dirac-Hara + a constant imaginary part 2 ground state + a constant imaginary part 3 Dirac-Hara + HL imag part + a constant imaginary part 5 Partially nonlocal: Dirac-Fock for core + HL for valence electrons + a constant imaginary part

#### Schema

```
|X| I F F I
```

### 16.36 feff.exelfs

|            |                                      |
|------------|--------------------------------------|
| Keyword    | feff.exelfs                          |
| Kind       | repeating-record                     |
| Type       | Repeating record with columns: Float |
| Default    | Required / No default                |
| Deprecated | No                                   |

This card is not implemented yet.

#### Schema

```
|X| F
```

```
|X| S
```

```
|X| ...
```

```
# In future, fix so that we actually have these params, not just strings.
```

### 16.37 feff.extpot

|         |             |
|---------|-------------|
| Keyword | feff.extpot |
| Kind    | scalar      |

|            |                       |
|------------|-----------------------|
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

Not currently working. Do not use.

## Schema

|X| L

### 16.38 feff.fms

|            |                                                      |
|------------|------------------------------------------------------|
| Keyword    | feff.fms                                             |
| Kind       | vector                                               |
| Type       | Vector: Float, Integer, Integer, Float, Float, Float |
| Default    | Required / No default                                |
| Deprecated | No                                                   |

feff.fms rfms lfms2 minv toler1 toler2 rdirec Use full multiple-scattering. rfms - radius of cluster to use for fms calculation. lfms1 - 0 for solids, 1 for molecules minv - set algorithm for matrix inversion 0: LU decomposition 2: Lanczos 3: Broyden (less reliable) toler1 - tolerance to stop recursion and Broyden algorithm toler2 - sets the matrix element of the Gt matrix to zero if its value is less than toler2 rdirec - sets the matrix element of the Gt matrix to zero if the distance between atoms is larger than rdirec

## Schema

|X| F I I F F F

### 16.39 feff.folp

|            |                                               |
|------------|-----------------------------------------------|
| Keyword    | feff.folp                                     |
| Kind       | repeating-record                              |
| Type       | Repeating record with columns: Integer, Float |
| Default    | Required / No default                         |
| Deprecated | No                                            |

feff.folp ipot overlap Set the overlap for unique potential ipot.

## Schema

|X| I F  
|X| ...

## 16.40 feff.fprime

|            |                             |
|------------|-----------------------------|
| Keyword    | feff.fprime                 |
| Kind       | vector                      |
| Type       | Vector: Float, Float, Float |
| Default    | Required / No default       |
| Deprecated | No                          |

feff.fprime emin emax estep emin - minimum energy of calculation. emax - maximum energy of calculation estep - energy step of calculation Calculate atomic scattering factor f'

### Schema

|X| F F F

## 16.41 feff.hole

|            |                        |
|------------|------------------------|
| Keyword    | feff.hole              |
| Kind       | vector                 |
| Type       | Vector: Integer, Float |
| Default    | Required / No default  |
| Deprecated | Yes                    |

DEPRECATED: Use feff.edge instead.

Specify the edge using the hole number ihole, e.g., K-edge : ihole = 1 L1-edge: ihole = 2 s02 specifies the EXAFS amplitude reduction factor, and should be set to 1.

### Schema

|X| I F

## 16.42 feff.hubbard

|            |                                      |
|------------|--------------------------------------|
| Keyword    | feff.hubbard                         |
| Kind       | vector                               |
| Type       | Vector: Float, Float, Float, Integer |
| Default    | Required / No default                |
| Deprecated | No                                   |

Not currently working.

### Schema

|X| F F F I

### 16.43 feff.interstitial

|            |                               |
|------------|-------------------------------|
| Keyword    | feff.interstitial             |
| Kind       | vector                        |
| Type       | Vector: Integer, Float, Float |
| Default    | Required / No default         |
| Deprecated | No                            |

feff.interstitial inters vtot vint\_scl The construction of the interstitial potential and density may be changed by using this card. inters = 1 might be useful when only the surroundings of the absorbing atom are specified in 'feff.inp'. inters defines how to find the interstitial potential. inters=0: The interstitial potential is found by averaging over the entire extended cluster in 'feff.inp'. (Default) inters=1: the interstitial potential is found locally around the absorbing atom. vtot: the volume per atom normalized by ratmin3 ( $vtot = (\text{volume per atom}) / \text{ratmin3}$ ), where ratmin is the shortest bond for the absorbing atom. This quantity defines the total volume (needed to calculate the interstitial density) of the extended cluster specified in 'feff.inp'. If  $vtot \leq 0$  then the total volume is calculated as a sum of Norman sphere volumes. Otherwise, total volume =  $nat * (vtot * \text{ratmin3})$ , where nat is the number of atoms in an extended cluster.

#### Schema

|X| I F F

### 16.44 feff.ion

|            |                                               |
|------------|-----------------------------------------------|
| Keyword    | feff.ion                                      |
| Kind       | repeating-record                              |
| Type       | Repeating record with columns: Integer, Float |
| Default    | Required / No default                         |
| Deprecated | No                                            |

feff.ion ipot ionization feff.ion ionizes all atoms with unique potential index ipot. NOTE: This input variable is not fully implemented yet.

#### Schema

|X| I F  
|X| ...

### 16.45 feff.iorder

|            |                       |
|------------|-----------------------|
| Keyword    | feff.iorder           |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.iorder iorder Set order of approximation when calculating effective scattering matrices.

## Schema

|X| I

### 16.46 feff.jumprm

|            |                       |
|------------|-----------------------|
| Keyword    | feff.jumprm           |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.jumprm UseRemoveJump If true, smooth the jump between muffin tin potentials and interstitial.

## Schema

|X| L

### 16.47 feff.kmesh

|            |                       |
|------------|-----------------------|
| Keyword    | feff.kmesh            |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

Specify the kmesh. feff.kmesh nkp(x) [nkpy nkpz [ktype [usesym] ] ] This card specifies the mesh of k-vectors used to sample the full Brillouin Zone for the evaluation of Brillouin Zone integrals. Nkp is the number of points used in the full zone. It can be specified either as "nkpx nkpy nkpz", "nkp", or "nkp 0 0". If usesym = 1, the zone is reduced to its irreducible wedge using the symmetry options specified in file symfile, which must be present in the working directory. The k-mesh is constructed using the tetrahedron method of Bloechl et al., Phys. Rev. B, 1990. The parameter ktype is meant for time-saving only and means: ktype=1 : regular mesh of nkp points for all modules ktype=2 : use nkp points for ldos/fms and nkp/5 points for pot (significant time savings) ktype=3 : use nkp points for ldos/fms and nkp/5 points for pot (near edge) ; reduce nkp for all modules as we get away from near-edge (somewhat experimental) \* use a k-mesh of 1000 points in the full BZ. feff.kmesh 1000 \* use a k-mesh of 10x5x3 points for a large, irregular cell. feff.kmesh 10 5 3 \* use a k-mesh of 1000 points and try to save time: feff.kmesh 1000 0 0 3

## Schema

|X| I ...

### 16.48 feff.lattice

|            |                                       |
|------------|---------------------------------------|
| Keyword    | feff.lattice                          |
| Kind       | repeating-record                      |
| Type       | Repeating record with columns: String |
| Default    | Required / No default                 |
| Deprecated | No                                    |

This card specifies the lattice. First, its type must be specified using a single letter : P for primitive, F for face centered cubic, I for body centered cubic, H for hexagonal. The following three lines give the three basis vectors in Carthesian Angstrom coordinates. They are multiplied by scale (e.g., 0.529177 to convert from bohr to Angstrom). feff.lattice P 3.18800 ax ay az bx by bz cx cy cz

#### Schema

```
# Need to fix this so that all parameters are set in separate fields.
|X| S
|X| ...
```

### 16.49 feff.ldec

|            |                       |
|------------|-----------------------|
| Keyword    | feff.ldec             |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

For use in calculations of NRIXS. Not implemented yet.

#### Schema

```
|X| I
```

### 16.50 feff.ldos

|            |                                      |
|------------|--------------------------------------|
| Keyword    | feff.ldos                            |
| Kind       | vector                               |
| Type       | Vector: Float, Float, Float, Integer |
| Default    | Required / No default                |
| Deprecated | No                                   |

feff.ldos emin emax broadening npoints Run angular momentum project density of states calculations. emin - minimum energy of energy grid. emax - maximum energy of energy grid. broadening - Lorentzian broadening to use in calculation. npoints - number of energy points in grid.

#### Schema

```
|X| F F F I
```

### 16.51 feff.lfms1

|            |            |
|------------|------------|
| Keyword    | feff.lfms1 |
| Kind       | scalar     |
| Type       | Integer    |
| Default    | -1         |
| Deprecated | No         |

set maximum angular momentum to use in potentials

#### Schema

-1 |X| I

### 16.52 feff.lfms2

|            |            |
|------------|------------|
| Keyword    | feff.lfms2 |
| Kind       | scalar     |
| Type       | Integer    |
| Default    | -1         |
| Deprecated | No         |

set maximum angular momentum to use in FMS

#### Schema

-1 |X| I

### 16.53 feff.ljmax

|            |                       |
|------------|-----------------------|
| Keyword    | feff.ljmax            |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

For use in calculations of NRIXS. Not implemented yet.

#### Schema

|X| I

### 16.54 feff.magic

|            |                       |
|------------|-----------------------|
| Keyword    | feff.magic            |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

This card is not implemented yet.

#### Schema

|X| F

### 16.55 feff.mbconv

|            |                       |
|------------|-----------------------|
| Keyword    | feff.mbconv           |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | Yes                   |

Deprecated: Should not be used.

#### Schema

|X| L

### 16.56 feff.mpse

|            |                          |
|------------|--------------------------|
| Keyword    | feff.mpse                |
| Kind       | vector                   |
| Type       | Vector: Integer, Integer |
| Default    | Required / No default    |
| Deprecated | No                       |

feff.mpse ipl npl Use many-pole model self-energy. This requires the loss function, which can be provided externally, or approximated using feff.opcons. ipl - Set method for self-energy inclusion: 1: use an "average" self-energy which is applied to the whole system (default). 2: use a density dependent self-energy which is different at every point inside the muffin-tin radius.

#### Schema

|X| I I

### 16.57 feff.multipole

|            |                          |
|------------|--------------------------|
| Keyword    | feff.multipole           |
| Kind       | vector                   |
| Type       | Vector: Integer, Integer |
| Default    | Required / No default    |
| Deprecated | No                       |

feff.multipole le2 l2lp Set multipole expansion approximation. Specifies which multipole transitions to include in the calculations. The options are: only dipole (le2 = 0, default), dipole and magnetic dipole (le2 = 1), dipole and quadrupole (le2 = 2). This cannot be used with NRIXS and is not supported with EXELFS and ELNES. The additional field l2lp can be used to calculate individual dipolar contributions coming from  $L \rightarrow L + 1$  (l2lp = 1) and from  $L \rightarrow L - 1$  (l2lp = -1). Notice that in polarization dependent data there is also a cross term, which is calculated only when l2lp = 0

#### Schema

|X| I I

### 16.58 feff.nleg

|            |                       |
|------------|-----------------------|
| Keyword    | feff.nleg             |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.nleg nleg Set maximum number of legs to use in path expansion. A Single scattering path has 2 legs.

#### Schema

|X| I

### 16.59 feff.nohole

|            |                       |
|------------|-----------------------|
| Keyword    | feff.nohole           |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | Yes                   |

feffnohole UseNoHole Do not use a hole in the calculation. DEPRECATED - Use feff.corehole instead.

## Schema

|X| L

### 16.60 feff.nrixs

|            |                                        |
|------------|----------------------------------------|
| Keyword    | feff.nrixs                             |
| Kind       | repeating-record                       |
| Type       | Repeating record with columns: Integer |
| Default    | Required / No default                  |
| Deprecated | No                                     |

## Schema

|X| I

|X| F ...

|X| ...

### 16.61 feff.nstar

|            |                       |
|------------|-----------------------|
| Keyword    | feff.nstar            |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.nstar WriteNStar Write nstar.dat with effective coordination number N\*.

## Schema

|X| L

### 16.62 feff.numdens

|            |                                               |
|------------|-----------------------------------------------|
| Keyword    | feff.numdens                                  |
| Kind       | repeating-record                              |
| Type       | Repeating record with columns: Integer, Float |
| Default    | Required / No default                         |
| Deprecated | No                                            |

## Schema

|X| I F

|X| ...

### 16.63 feff.opcons

|            |                       |
|------------|-----------------------|
| Keyword    | feff.opcons           |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

Calculate the dielectric function of the material within the atomic approximation. Very fast. Used to calculate loss.dat for use with feff.mpse self-energy calculations.

#### Schema

|X| L

### 16.64 feff.overlap

|            |                       |
|------------|-----------------------|
| Keyword    | feff.overlap          |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.overlap can be used to construct approximate overlapped atom potentials when atomic coordinates are not known or specified. NOTE: This input variable is not fully implemented yet.

#### Schema

|X| I

### 16.65 feff.pcriteria

|            |                       |
|------------|-----------------------|
| Keyword    | feff.pcriteria        |
| Kind       | vector                |
| Type       | Vector: Float, Float  |
| Default    | Required / No default |
| Deprecated | No                    |

feff.pcriteria pcrith pcrith Set criteria for filtering in pathfinder. The keep-criterion pcrith looks at the amplitude of chi (in the plane wave approximation) for the current path and compares it to a single scattering path of the same effective length. To set this value, consider the maximum degeneracy you expect and divide your plane wave criterion by this number. For example, in fcc Cu, typical degeneracies are 196 for paths with large r, and the minimum degeneracy is 6. So a keep criterion of 0.08heap-criterion pcrith filters paths as the pathfinder puts all paths into a heap (a partially ordered data structure), then removes them in order of increasing total path length. Each path that is removed from the heap is modified and then considered again as part of the search algorithm. The heap filter is used to decide if a path has enough amplitude in it to be worth

further consideration. If a path can be eliminated at this point, entire trees of derivative paths can be neglected, leading to enormous time savings. This test does not come into play until paths with at least 4 legs are being considered, so single scattering and triangular (2 and 3 legged) paths will always pass this test. Because only a small part of a path is used for this criterion, it is difficult to predict what appropriate values will be.

## Schema

|X| F F

### 16.66 feff.pmbse

|            |                                            |
|------------|--------------------------------------------|
| Keyword    | feff.pmbse                                 |
| Kind       | vector                                     |
| Type       | Vector: Integer, Integer, Integer, Integer |
| Default    | Required / No default                      |
| Deprecated | No                                         |

## Schema

|X| I I I I

### 16.67 feff.polarization

|            |                             |
|------------|-----------------------------|
| Keyword    | feff.polarization           |
| Kind       | vector                      |
| Type       | Vector: Float, Float, Float |
| Default    | Required / No default       |
| Deprecated | No                          |

feff.polarization x y z Set polarization vector of x-rays in cartesian coordinates of input file.

## Schema

|X| F F F

### 16.68 feff.potentials

|            |                                                                                  |
|------------|----------------------------------------------------------------------------------|
| Keyword    | feff.potentials                                                                  |
| Kind       | repeating-record                                                                 |
| Type       | Repeating record with columns: Integer, Integer, String, Integer, Integer, Float |
| Default    | Required / No default                                                            |
| Deprecated | No                                                                               |

feff.potentials ipot iz pot\_label lfms1 lfms2 stoichiometry ... Set unique potentials for FEFF calculation.

## Schema

```
|X| I I S I I F
|X| ...
```

### 16.69 feff.potentials.spin

|            |                                               |
|------------|-----------------------------------------------|
| Keyword    | feff.potentials.spin                          |
| Kind       | repeating-record                              |
| Type       | Repeating record with columns: Integer, Float |
| Default    | Required / No default                         |
| Deprecated | No                                            |

feff.potentials.spin ipot spin\_moment ... Set spin moments of atoms defined in potentials card

## Schema

```
|X| I F
|X| ...
```

### 16.70 feff.preps

|            |                       |
|------------|-----------------------|
| Keyword    | feff.preps            |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.preps PrintEpsilon Print dielectric function as calculated by feff.opcons.

## Schema

```
|X| L
```

### 16.71 feff.print

|            |                                                              |
|------------|--------------------------------------------------------------|
| Keyword    | feff.print                                                   |
| Kind       | vector                                                       |
| Type       | Vector: Integer, Integer, Integer, Integer, Integer, Integer |
| Default    | Required / No default                                        |
| Deprecated | No                                                           |

feff.pring ipr\_pot ipr\_xsph ipr\_fms ipr\_path ipr\_genfmt ipr\_ff2x Set print level for each module of FEFF: pot: 0 write 'pot.bin' only 1 add 'misc.dat' 2 add 'potNN.dat' 3 add 'atomNN.dat' xsph: 0 write 'phase.bin' and 'xsect.bin' only 1 add 'axafs.dat' and 'phase.dat' 2 add 'phaseNN.dat' and 'phminNN.dat' 3 add 'ratio.dat' (for XMCD normalization) and 'emesh.dat'. fms: 0 write 'gg.bin' 1 write 'gg.dat'

path: 0 write 'paths.dat' only 1 add 'crit.dat' 3 add 'fbeta' files (plane wave  $|f(\beta)|$  approximations) 5 Write only 'crit.dat' and do not write 'paths.dat'. (This is useful genfmt 0 write 'list.dat', and write 'feff.bin' with all paths with importance greater than or equal to two thirds of the curved wave importance criterion

genfmt: 0 write 'list.dat', and write 'feff.bin' with all paths with importance greater than or equal to two thirds of the curved wave importance criterion 1 write all paths to 'feff.bin'

ff2x: 0 write 'chi.dat' and 'xmu.dat' 2 add 'chipNNNN.dat' (chi(k) for each path individually) 3 add 'feffNNNN.dat' and 'files.dat', and do not add 'chipNNNN.dat' files

## Schema

|X| I I I I I I

### 16.72 feff.rconv

|            |                       |
|------------|-----------------------|
| Keyword    | feff.rconv            |
| Kind       | vector                |
| Type       | Vector: Float, String |
| Default    | Required / No default |
| Deprecated | No                    |

Advanced: Print running convolution with the spectral function.

## Schema

|X| F S

### 16.73 feff.real

|            |                       |
|------------|-----------------------|
| Keyword    | feff.real             |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

Advanced: Use real phase shifts.

## Schema

|X| L

### 16.74 feff.reciprocal

|         |                 |
|---------|-----------------|
| Keyword | feff.reciprocal |
| Kind    | scalar          |
| Type    | Logical         |

|            |                       |
|------------|-----------------------|
| Default    | Required / No default |
| Deprecated | No                    |

Work in reciprocal space (k-space calculation).

#### Schema

|X| L

### 16.75 feff.restart

|            |                       |
|------------|-----------------------|
| Keyword    | feff.restart          |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

Not implemented yet.

#### Schema

|X| L

### 16.76 feff.rgrid

|            |                       |
|------------|-----------------------|
| Keyword    | feff.rgrid            |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| F

### 16.77 feff.rhozzp

|            |                       |
|------------|-----------------------|
| Keyword    | feff.rhozzp           |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

Not implemented yet.

## Schema

|X| L

### 16.78 feff.rixs

|            |                             |
|------------|-----------------------------|
| Keyword    | feff.rixs                   |
| Kind       | vector                      |
| Type       | Vector: Float, Float, Float |
| Default    | Required / No default       |
| Deprecated | No                          |

feff.RIXS gam\_Ei gam\_El xmu The RIXS card sets the parameters for the RIXS calculation. It must be used with a workflow or script to produce RIXS. gam\_Ei Broadening to use in the incident energy direction.

gam\_El Broadening to use in the energy loss direction.

xmu Fermi energy to use for the RIXS calculation, if not present, the fermi level calculated in the SCF step will be used.

## Schema

|X| F F F

### 16.79 feff.rlprint

|            |                       |
|------------|-----------------------|
| Keyword    | feff.rlprint          |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

Advanced used in RIXS calculations, but automated by corvus. Prints real scattering wavefunction  $R_l$ .

## Schema

|X| L

### 16.80 feff.rmultiplier

|            |                       |
|------------|-----------------------|
| Keyword    | feff.rmultiplier      |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

feff.rmultiplier rmult Multiply coordinates of all atoms by rmult, expanding or contracting the system.

## Schema

|X| F

### 16.81 feff.rpath

|            |                       |
|------------|-----------------------|
| Keyword    | feff.rpath            |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

feff.rpath rmax Set maximum path length for path expansion calculations of EXAFS, EXELFS, DAFS, etc.

## Schema

|X| F

### 16.82 feff.rphases

|            |                       |
|------------|-----------------------|
| Keyword    | feff.rphases          |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.rphases UseRealPhases Set real phase shift approximation. UseRealPhases - use real phase shift if true.

## Schema

|X| L

### 16.83 feff.rsigma

|            |                       |
|------------|-----------------------|
| Keyword    | feff.rsigma           |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.rsigma UseRealSelfEnergy If true use only real part of self-energy.

## Schema

|X| L

## 16.84 feff.s02

|            |                       |
|------------|-----------------------|
| Keyword    | feff.s02              |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

feff.s02 s02 Set EXAFS amplitude reduction factor.

### Schema

|X| F

## 16.85 feff.scf

|            |                                                 |
|------------|-------------------------------------------------|
| Keyword    | feff.scf                                        |
| Kind       | vector                                          |
| Type       | Vector: Float, Integer, Integer, Float, Integer |
| Default    | Required / No default                           |
| Deprecated | No                                              |

feff.scf rfms1 lfms1 nscmt ca nmix Use self-consistent field calculation of potentials. rfms1 - radius of cluster to use for scf calculations. lfms1 - 0 for solids, 1 for molecules nscmt - maximum number of iterations in SCF calculation. ca - convergence factor for Broyden algorithm nmix - number of initial iterations of mixing algorithm to use.

### Schema

|X| F I I F I

## 16.86 feff.scframp

|            |                        |
|------------|------------------------|
| Keyword    | feff.scframp           |
| Kind       | vector                 |
| Type       | Vector: Float, Integer |
| Default    | Required / No default  |
| Deprecated | No                     |

Ramp scf radius up to final radius, starting from start\_radius, and taking n\_ramp steps.  
SCFRAMP start\_radius n\_ramp

### Schema

|X| F I

## 16.87 feff.screen

|            |                                               |
|------------|-----------------------------------------------|
| Keyword    | feff.screen                                   |
| Kind       | repeating-record                              |
| Type       | Repeating record with columns: String, String |
| Default    | Required / No default                         |
| Deprecated | No                                            |

The screen module, which calculates the RPA core hole potential, is a 'silent' module: it has no obvious input but instead runs entirely on default values. Using the feff.SCREEN card you can change these default values. They will be written to an optional 'screen.inp' file (which you can also edit manually). The feff.SCREEN card can occur more than once in 'feff.inp'; all entries will be applied to the calculation. parameter must be one of : ner (40), nei (20), maxl (4), irrh (1), iend (0), lfx (0), emin (-40 eV), emax (0 eV), eimax (2 eV), ermin (0.001 eV), rfms (4.0), nrptx0 (251). For most calculations the default values given (between brackets) are fine. Occasionally we've changed rfms, maxl, or emin. Note that the screen is only active with feff.COREHOLE RPA . \* Set the cluster radius for the RPA potential calculation higher than the default of 4.0 feff.COREHOLE RPA feff.SCREEN rfms 5.0

### Schema

```
# We will also want to improve this interface.
|X| S S
|X| ...
```

## 16.88 feff.scxc

|            |                       |
|------------|-----------------------|
| Keyword    | feff.scxc             |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

### Schema

```
|X| I
```

## 16.89 feff.self

|            |                       |
|------------|-----------------------|
| Keyword    | feff.self             |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.self PrintSelfEnergy Print out quasiparticle self-energy calculated during spectral-function convolution calculations.

## Schema

|X| L

### 16.90 feff.setedge

|            |                       |
|------------|-----------------------|
| Keyword    | feff.setedge          |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.setedge UseEdgeTable Use table of experimental edge energies.

## Schema

|X| L

### 16.91 feff.sfconv

|            |                       |
|------------|-----------------------|
| Keyword    | feff.sfconv           |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.sfconv UseSpectralFunctionConvolution If true, convolve spectrum with spectral function to account for multi-electron excitations.

## Schema

|X| L

### 16.92 feff.sfse

|            |                       |
|------------|-----------------------|
| Keyword    | feff.sfse             |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

feff.sfse k Print out off shell self-energy  $\Sigma(k,E)$  calculated during spectral-function convolution calculations.

## Schema

|X| F

### 16.93 feff.sgroup

|            |                       |
|------------|-----------------------|
| Keyword    | feff.sgroup           |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

Specify space group by number (1 - 230).

#### Schema

|X| L

### 16.94 feff.sig2

|            |                       |
|------------|-----------------------|
| Keyword    | feff.sig2             |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

feff.sig2 sig2 Set a single Debye-Waller factor for all paths,  $\exp(-2 \cdot \text{sig2} \cdot k^2)$ .

#### Schema

|X| F

### 16.95 feff.sig3

|            |                       |
|------------|-----------------------|
| Keyword    | feff.sig3             |
| Kind       | vector                |
| Type       | Vector: Float, Float  |
| Default    | Required / No default |
| Deprecated | No                    |

feff.sig3 alphas thetas Set first and third cumulants for single scattering paths: alphas - first cumulant thetas - third cumulant

#### Schema

|X| F F

## 16.96 feff.siggk

|            |                       |
|------------|-----------------------|
| Keyword    | feff.siggk            |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

Not implemented yet.

### Schema

|X| F

## 16.97 feff.spin

|            |                                      |
|------------|--------------------------------------|
| Keyword    | feff.spin                            |
| Kind       | vector                               |
| Type       | Vector: Integer, Float, Float, Float |
| Default    | Required / No default                |
| Deprecated | No                                   |

feff.spin ispin sx sy sz Specify the type of spin-dependent calculation for spin along the (x, y, z) direction, along the z-axis by default. The SPIN card is required for the calculation of all spin-dependent effects, including XMCD and SPXAS. ispin: 2 spin-up SPXAS and LDOS -2 spin-down SPXAS and LDOS 1 spin-up portion of XMCD calculations -1 spin-down portion of XMCD calculations

### Schema

|X| I F F F

## 16.98 feff.ss

|            |                                                               |
|------------|---------------------------------------------------------------|
| Keyword    | feff.ss                                                       |
| Kind       | repeating-record                                              |
| Type       | Repeating record with columns: Integer, Integer, Float, Float |
| Default    | Required / No default                                         |
| Deprecated | No                                                            |

Set an

### Schema

|X| I I F F  
|X| ...

## 16.99 feff.strfac

|            |                             |
|------------|-----------------------------|
| Keyword    | feff.strfac                 |
| Kind       | vector                      |
| Type       | Vector: Float, Float, Float |
| Default    | Required / No default       |
| Deprecated | No                          |

feff.strfac eta gmax rmax This card gives three non-physical internal parameters for the calculation of the KKR structure factors : the Ewald parameter and a multiplicative cutoff factor for sums over reciprocal (gmax) and real space (rmax) sums. Multiplicative means the code makes a 'smart' guess of a cutoff radius, but if one suspects something fishy is going on, one can here e.g. use gmax=2 to multiply this guess by 2. Eta is an absolute number. Given the stability of the Ewald algorithm, it shouldn't be necessary to use this card. Its use is not recommended. Only active in combination with the feff.reciprocal card.

### Schema

|X| F F F

## 16.100 feff.symmetry

|            |                       |
|------------|-----------------------|
| Keyword    | feff.symmetry         |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.symmetry UseSymmetry If false, turn off symmetry considerations in path expansion.

### Schema

|X| L

## 16.101 feff.target

|            |                       |
|------------|-----------------------|
| Keyword    | feff.target           |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.target ic Specifies the location of the absorber atom for reciprocal space calculations. It is entry ic of the feff.ATOMS card if an feff.ATOMS card and feff.LATTICE card are used. In conjunction with the cif\_input card it is entry ic the list of atoms as given in the '.cif' file (i.e., a list of the crystallographically inequivalent atom positions in the unit cell). The target needs to be specified also for NOHOLE calculations. Note that this cannot be specified in the feff.POTENTIALS list because periodic boundary conditions would then produce an infinite number of core holes. \* calculate a spectrum for the second atom in the feff.ATOMS list or CIF file. feff.TARGET 2

## Schema

|X| I

### 16.102 feff.tdlda

|            |                       |
|------------|-----------------------|
| Keyword    | feff.tdlda            |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.tdlda ifxc Use TDLDA to calculate x-ray absorption spectrum. ifxc - set algorithm for tdlda approximation 0: use static approximation for screening of the x-ray field 1: use approximate dynamic screening of x-ray field and core-hole.

## Schema

|X| I

### 16.103 feff.temperature

|            |                        |
|------------|------------------------|
| Keyword    | feff.temperature       |
| Kind       | vector                 |
| Type       | Vector: Float, Integer |
| Default    | Required / No default  |
| Deprecated | No                     |

feff.temperature etemp iscfxc The TEMP card sets the electronic temperature and the exchange-correlation potential. etemp = 0 (default). etemp is in eV. There are 4 different options for the exchange-correlation potential. iscfxc No temperature dependent exchange correlations 11 von-Barth Hedin 1971 (default) 12 Perdew-Zunger Explicitly temperature dependent exchange correlations 21 Perrot Dharma-Wardana 1984 22 KSDT (recommended)

## Schema

|X| F I

### 16.104 feff.title

|            |                                       |
|------------|---------------------------------------|
| Keyword    | feff.title                            |
| Kind       | repeating-record                      |
| Type       | Repeating record with columns: String |
| Default    | Required / No default                 |
| Deprecated | No                                    |

Set title for this feff calculation.

## Schema

|X| S ...  
|X| ...

### 16.105 feff.tolscf

|            |                             |
|------------|-----------------------------|
| Keyword    | feff.tolscf                 |
| Kind       | vector                      |
| Type       | Vector: Float, Float, Float |
| Default    | Required / No default       |
| Deprecated | No                          |

set tolerances for scf convergence feff.tolscf fermi\_tol charge\_tol partial\_charge\_tol

## Schema

|X| F F F

### 16.106 feff.unfreezef

|            |                       |
|------------|-----------------------|
| Keyword    | feff.unfreezef        |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.unfreezef UnfreezeFOrbitals If true, allow f-orbitals to relax during SCF calculation.

## Schema

|X| L

### 16.107 feff.xanes

|            |                       |
|------------|-----------------------|
| Keyword    | feff.xanes            |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

feff.xanes xkmax Calculate XANES spectrum. xkmax - calculate up to  $k = xkmax$

## Schema

|X| F

### 16.108 feff.xes

|            |                             |
|------------|-----------------------------|
| Keyword    | feff.xes                    |
| Kind       | vector                      |
| Type       | Vector: Float, Float, Float |
| Default    | Required / No default       |
| Deprecated | No                          |

feff.xes emin emax estep Calculate x-ray emission spectrum. emin - minimum energy of calculation. emax - maximum energy of calculation estep - energy step of calculation

#### Schema

|X| F F F

### 16.109 feff.xmcd

|            |                             |
|------------|-----------------------------|
| Keyword    | feff.xmcd                   |
| Kind       | vector                      |
| Type       | Vector: Float, Float, Float |
| Default    | Required / No default       |
| Deprecated | No                          |

feff.xmcd xkmax xkstep estep Calculate x-ray magnetic (or natural) circular dichroism xkmax - calculate up to  $k = xkmax$  xkstep - use steps of size xkstep estep - near the edge, use steps calculated from estep

#### Schema

|X| F F F

### 16.110 nuctemp

|            |         |
|------------|---------|
| Keyword    | nuctemp |
| Kind       | scalar  |
| Type       | Float   |
| Default    | 300.0   |
| Deprecated | No      |

nuctemp temp

Specify the temperature (in K) for the nuclear motion (DW factors).

#### Schema

300.0 |X| F

### 16.111 opcons.drude

|            |                       |
|------------|-----------------------|
| Keyword    | opcons.drude          |
| Kind       | vector                |
| Type       | Vector: Float, Float  |
| Default    | Required / No default |
| Deprecated | No                    |

Calculate a drude approximation for the low energy part of the spectrum in opcons.

#### Schema

|X| F F

### 16.112 opcons.usesaved

|            |                       |
|------------|-----------------------|
| Keyword    | opcons.usesaved       |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

feff.opcons Calculate loss function using an atomic approximation.

#### Schema

|X| L

### 16.113 run\_dmdw

|            |          |
|------------|----------|
| Keyword    | run_dmdw |
| Kind       | scalar   |
| Type       | Logical  |
| Default    | False    |
| Deprecated | No       |

run standalone dmdw along with feff runs.

#### Schema

False |X| L

## 17 ORCA

### 17.1 orca.basis.basis

|            |                  |
|------------|------------------|
| Keyword    | orca.basis.basis |
| Kind       | scalar           |
| Type       | String           |
| Default    | def2-TZVP        |
| Deprecated | No               |

#### Schema

def2-TZVP |X| S

### 17.2 orca.basis.decontract

|            |                       |
|------------|-----------------------|
| Keyword    | orca.basis.decontract |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| L

### 17.3 orca.ci

|            |         |
|------------|---------|
| Keyword    | orca.ci |
| Kind       | scalar  |
| Type       | Logical |
| Default    | F       |
| Deprecated | No      |

#### Schema

F |X| L

### 17.4 orca.citype

|            |             |
|------------|-------------|
| Keyword    | orca.citype |
| Kind       | scalar      |
| Type       | String      |
| Default    | CCSD        |
| Deprecated | No          |

#### Schema

CCSD |X| S

## 17.5 orca.coords.units

|            |                       |
|------------|-----------------------|
| Keyword    | orca.coords.units     |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

### Schema

|X| S

## 17.6 orca.cpcm

|            |                       |
|------------|-----------------------|
| Keyword    | orca.cpcm             |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

### Schema

|X| S

## 17.7 orca.expert

|            |                                       |
|------------|---------------------------------------|
| Keyword    | orca.expert                           |
| Kind       | repeating-record                      |
| Type       | Repeating record with columns: String |
| Default    | Required / No default                 |
| Deprecated | No                                    |

Use verbatim expert setting for orca.

### Schema

|X| S ...

|X| ...

## 17.8 orca.geom.optimizationquality

|         |                               |
|---------|-------------------------------|
| Keyword | orca.geom.optimizationquality |
| Kind    | scalar                        |
| Type    | String                        |
| Default | TIGHTOPT                      |

|            |    |
|------------|----|
| Deprecated | No |
|------------|----|

#### Schema

TIGHTOPT |X| S

### 17.9 orca.method.allowrhf

|            |                      |
|------------|----------------------|
| Keyword    | orca.method.allowrhf |
| Kind       | scalar               |
| Type       | Logical              |
| Default    | F                    |
| Deprecated | No                   |

#### Schema

F |X| L

### 17.10 orca.method.amass

|            |                   |
|------------|-------------------|
| Keyword    | orca.method.amass |
| Kind       | scalar            |
| Type       | String            |
| Default    | Mass2016          |
| Deprecated | No                |

#### Schema

Mass2016 |X| S

### 17.11 orca.method.frozenscore

|            |                         |
|------------|-------------------------|
| Keyword    | orca.method.frozenscore |
| Kind       | scalar                  |
| Type       | Logical                 |
| Default    | F                       |
| Deprecated | No                      |

#### Schema

F |X| L

### 17.12 orca.method.grid

|            |                  |
|------------|------------------|
| Keyword    | orca.method.grid |
| Kind       | scalar           |
| Type       | String           |
| Default    | GRID4            |
| Deprecated | No               |

#### Schema

GRID4 |X| S

### 17.13 orca.method.gridx

|            |                       |
|------------|-----------------------|
| Keyword    | orca.method.gridx     |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| S

### 17.14 orca.method.method

|            |                    |
|------------|--------------------|
| Keyword    | orca.method.method |
| Kind       | scalar             |
| Type       | String             |
| Default    | B3LYP              |
| Deprecated | No                 |

#### Schema

B3LYP |X| S

### 17.15 orca.method.ri

|            |                |
|------------|----------------|
| Keyword    | orca.method.ri |
| Kind       | scalar         |
| Type       | Logical        |
| Default    | F              |
| Deprecated | No             |

#### Schema

F |X| L

### 17.16 orca.method.runtype

|            |                     |
|------------|---------------------|
| Keyword    | orca.method.runtype |
| Kind       | scalar              |
| Type       | String              |
| Default    | OPT                 |
| Deprecated | No                  |

#### Schema

OPT |X| S

### 17.17 orca.method.usesymm

|            |                     |
|------------|---------------------|
| Keyword    | orca.method.usesymm |
| Kind       | scalar              |
| Type       | Logical             |
| Default    | F                   |
| Deprecated | No                  |

#### Schema

F |X| L

### 17.18 orca.mp2

|            |          |
|------------|----------|
| Keyword    | orca.mp2 |
| Kind       | scalar   |
| Type       | Logical  |
| Default    | F        |
| Deprecated | No       |

#### Schema

F |X| L

### 17.19 orca.mp2type

|            |              |
|------------|--------------|
| Keyword    | orca.mp2type |
| Kind       | scalar       |
| Type       | String       |
| Default    | MP2          |
| Deprecated | No           |

#### Schema

MP2 |X| S

### 17.20 orca.output.pdbfile

|            |                       |
|------------|-----------------------|
| Keyword    | orca.output.pdbfile   |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| L

### 17.21 orca.output.print

|            |                       |
|------------|-----------------------|
| Keyword    | orca.output.print     |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| S

### 17.22 orca.output.printlevel

|            |                        |
|------------|------------------------|
| Keyword    | orca.output.printlevel |
| Kind       | scalar                 |
| Type       | String                 |
| Default    | Required / No default  |
| Deprecated | No                     |

#### Schema

|X| S

### 17.23 orca.output.xyzfile

|            |                       |
|------------|-----------------------|
| Keyword    | orca.output.xyzfile   |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| L

### 17.24 orca.rel.method

|            |                       |
|------------|-----------------------|
| Keyword    | orca.rel.method       |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| S

### 17.25 orca.rel.soctype

|            |                       |
|------------|-----------------------|
| Keyword    | orca.rel.soctype      |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| S

### 17.26 orca.scf.autostart

|            |                    |
|------------|--------------------|
| Keyword    | orca.scf.autostart |
| Kind       | scalar             |
| Type       | Logical            |
| Default    | T                  |
| Deprecated | No                 |

#### Schema

T |X| L

### 17.27 orca.scf.cnvdamp

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.cnvdamp      |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| L

### 17.28 orca.scf.cnvshift

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.cnvshift     |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| L

### 17.29 orca.scf.convergence

|            |                      |
|------------|----------------------|
| Keyword    | orca.scf.convergence |
| Kind       | scalar               |
| Type       | String               |
| Default    | NORMALSCF            |
| Deprecated | No                   |

#### Schema

NORMALSCF |X| S

### 17.30 orca.scf.convergencestrategy

|            |                              |
|------------|------------------------------|
| Keyword    | orca.scf.convergencestrategy |
| Kind       | scalar                       |
| Type       | String                       |
| Default    | NormalConv                   |
| Deprecated | No                           |

#### Schema

NormalConv |X| S

### 17.31 orca.scf.diis

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.diis         |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| L

### 17.32 orca.scf.fracocc

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.fracocc      |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| S

### 17.33 orca.scf.guess

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.guess        |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| S

### 17.34 orca.scf.hftyp

|            |                |
|------------|----------------|
| Keyword    | orca.scf.hftyp |
| Kind       | scalar         |
| Type       | String         |
| Default    | RHF            |
| Deprecated | No             |

#### Schema

RHF |X| S

### 17.35 orca.scf.jmatrix

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.jmatrix      |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| S

### 17.36 orca.scf.kdiis

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.kdiis        |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| L

### 17.37 orca.scf.keepdens

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.keepdens     |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| L

### 17.38 orca.scf.keepints

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.keepints     |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| L

### 17.39 orca.scf.kmatrix

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.kmatrix      |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| S

#### 17.40 orca.scf.maxiter

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.maxiter      |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| S

#### 17.41 orca.scf.nr

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.nr           |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| L

#### 17.42 orca.scf.readints

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.readints     |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| L

#### 17.43 orca.scf.scfmode

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.scfmode      |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| S

#### 17.44 orca.scf.smeartemp

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.smeartemp    |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

##### Schema

|X| L

#### 17.45 orca.scf.soscf

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.soscf        |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

##### Schema

|X| L

#### 17.46 orca.scf.uno

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.uno          |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

##### Schema

|X| L

#### 17.47 orca.scf.usecheapints

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.usecheapints |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

##### Schema

|X| L

### 17.48 orca.scf.valformat

|            |                       |
|------------|-----------------------|
| Keyword    | orca.scf.valformat    |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| S

### 17.49 orca.tole

|            |                       |
|------------|-----------------------|
| Keyword    | orca.tole             |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| F

### 17.50 orca.tolmaxg

|            |                       |
|------------|-----------------------|
| Keyword    | orca.tolmaxg          |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| F

### 17.51 orca.tolrmaxd

|            |                       |
|------------|-----------------------|
| Keyword    | orca.tolrmaxd         |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

|X| F

## 17.52 orca.tolrmsd

|            |                       |
|------------|-----------------------|
| Keyword    | orca.tolrmsd          |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

### Schema

|X| F

## 17.53 orca.tolrmsg

|            |                       |
|------------|-----------------------|
| Keyword    | orca.tolrmsg          |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

### Schema

|X| F

## 18 SIESTA

### 18.1 phsf.broadening

|            |                 |
|------------|-----------------|
| Keyword    | phsf.broadening |
| Kind       | scalar          |
| Type       | Float           |
| Default    | 0.5             |
| Deprecated | No              |

### Schema

0.5 |X| F

### 18.2 phsf.ekeqp

|            |                      |
|------------|----------------------|
| Keyword    | phsf.ekeqp           |
| Kind       | vector               |
| Type       | Vector: Float, Float |
| Default    | 0.0 0.0              |
| Deprecated | No                   |

## Schema

0.0 0.0 |X| F F

### 18.3 phsf.numtimepoints

|            |                    |
|------------|--------------------|
| Keyword    | phsf.numtimepoints |
| Kind       | scalar             |
| Type       | Integer            |
| Default    | 40000              |
| Deprecated | No                 |

## Schema

40000 |X| I

### 18.4 siesta.AtomicCoordinatesFormat

|            |                                |
|------------|--------------------------------|
| Keyword    | siesta.AtomicCoordinatesFormat |
| Kind       | scalar                         |
| Type       | String                         |
| Default    | Fractional                     |
| Deprecated | No                             |

Specify units and format of atomic coordinates. Borh - cartesian in Bohr. Ang - cartesian in Angstrom. ScaledCartesian - cartesian scaled by lattice constant. Fractional - coordinates are given in units of the lattice vectors.

## Schema

Fractional |X| S

### 18.5 siesta.Beta.Broadening

|            |                        |
|------------|------------------------|
| Keyword    | siesta.Beta.Broadening |
| Kind       | scalar                 |
| Type       | Float                  |
| Default    | 0.1                    |
| Deprecated | No                     |

## Schema

0.1 |X| F

### 18.6 siesta.Block.AtomicCoordinatesAndAtomicSpecies

|            |                                                             |
|------------|-------------------------------------------------------------|
| Keyword    | siesta.Block.AtomicCoordinatesAndAtomicSpecies              |
| Kind       | repeating-record                                            |
| Type       | Repeating record with columns: Float, Float, Float, Integer |
| Default    | Required / No default                                       |
| Deprecated | No                                                          |

Specify coordinates of each atom as well as the species index. x1 y1 z1 SpeciesIndex1 x2 y2 z2 SpeciesIndex2 ...

#### Schema

```
|X| F F F I
|X| ...
```

### 18.7 siesta.Block.ChemicalSpeciesLabel

|            |                                   |
|------------|-----------------------------------|
| Keyword    | siesta.Block.ChemicalSpeciesLabel |
| Kind       | vector                            |
| Type       | Vector: Integer, Integer, String  |
| Default    | Required / No default             |
| Deprecated | No                                |

Indicate the different chemical species in the unit cell. Format: SpeciesIndex1 AtomicNumber1 AtomicSymbol1 SpeciesIndex2 AtomicNumber2 AtomicSymbol2 ...

#### Schema

```
|X| I I S
```

### 18.8 siesta.Block.LatticeParameters

|            |                                                  |
|------------|--------------------------------------------------|
| Keyword    | siesta.Block.LatticeParameters                   |
| Kind       | vector                                           |
| Type       | Vector: Float, Float, Float, Float, Float, Float |
| Default    | Required / No default                            |
| Deprecated | No                                               |

Set lattice parameters, i.e., a b c alpha beta gamma

#### Schema

```
|X| F F F F F F
```

## 18.9 siesta.Block.PAO.Basis

|            |                                       |
|------------|---------------------------------------|
| Keyword    | siesta.Block.PAO.Basis                |
| Kind       | repeating-record                      |
| Type       | Repeating record with columns: String |
| Default    | Required / No default                 |
| Deprecated | No                                    |

Set siesta basis manually

### Schema

|X| S ...  
|X| ...

## 18.10 siesta.Coreresponse.Broadening

|            |                                |
|------------|--------------------------------|
| Keyword    | siesta.Coreresponse.Broadening |
| Kind       | scalar                         |
| Type       | Float                          |
| Default    | 0.5                            |
| Deprecated | No                             |

### Schema

0.5 |X| F

## 18.11 siesta.Diag.DivideAndConquer

|            |                              |
|------------|------------------------------|
| Keyword    | siesta.Diag.DivideAndConquer |
| Kind       | scalar                       |
| Type       | String                       |
| Default    | Required / No default        |
| Deprecated | No                           |

### Schema

|X| S

## 18.12 siesta.DM.InitSpin.AF

|         |                       |
|---------|-----------------------|
| Keyword | siesta.DM.InitSpin.AF |
| Kind    | scalar                |
| Type    | Logical               |
| Default | Required / No default |

|            |    |
|------------|----|
| Deprecated | No |
|------------|----|

Set magnetic behavior of system to antiferromagnetic if true. Default is false and will be set up as ferromagnetic.

## Schema

|X| L

### 18.13 siesta.DM.MixingWeight

|            |                        |
|------------|------------------------|
| Keyword    | siesta.DM.MixingWeight |
| Kind       | scalar                 |
| Type       | Float                  |
| Default    | 0.10                   |
| Deprecated | No                     |

Mixing weight for the density matrix.

## Schema

0.10 |X| F

### 18.14 siesta.DM.NumberBroyden

|            |                         |
|------------|-------------------------|
| Keyword    | siesta.DM.NumberBroyden |
| Kind       | scalar                  |
| Type       | Integer                 |
| Default    | Required / No default   |
| Deprecated | No                      |

Set number of broyden steps

## Schema

|X| I

### 18.15 siesta.DM.NumberBroyden

|            |                         |
|------------|-------------------------|
| Keyword    | siesta.DM.NumberBroyden |
| Kind       | scalar                  |
| Type       | Integer                 |
| Default    | Required / No default   |
| Deprecated | No                      |

## Schema

|X| I

### 18.16 siesta.DM.NumberPulay

|            |                       |
|------------|-----------------------|
| Keyword    | siesta.DM.NumberPulay |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

Set number of Pulay steps

## Schema

|X| I

### 18.17 siesta.DM.Tolerance

|            |                     |
|------------|---------------------|
| Keyword    | siesta.DM.Tolerance |
| Kind       | scalar              |
| Type       | Float               |
| Default    | 1.0e-4              |
| Deprecated | No                  |

Convergence tolerance for elements of the density matrix

## Schema

1.0e-4 |X| F

### 18.18 siesta.ElectronicTemperature

|            |                              |
|------------|------------------------------|
| Keyword    | siesta.ElectronicTemperature |
| Kind       | vector                       |
| Type       | Vector: Float, String        |
| Default    | 1.0 K                        |
| Deprecated | No                           |

Set electronic teperature used to define occupation numbers during SCF.

## Schema

1.0 K |X| F S

### 18.19 siesta.expert

|            |                                       |
|------------|---------------------------------------|
| Keyword    | siesta.expert                         |
| Kind       | repeating-record                      |
| Type       | Repeating record with columns: String |
| Default    | Required / No default                 |
| Deprecated | No                                    |

#### Schema

|X| S ...  
|X| ...

### 18.20 siesta.LatticeConstant

|            |                        |
|------------|------------------------|
| Keyword    | siesta.LatticeConstant |
| Kind       | vector                 |
| Type       | Vector: Float, String  |
| Default    | Required / No default  |
| Deprecated | No                     |

Set an overall multiplicative constant for the lattice parameters.

#### Schema

|X| F S

### 18.21 siesta.LongOutput

|            |                   |
|------------|-------------------|
| Keyword    | siesta.LongOutput |
| Kind       | scalar            |
| Type       | Logical           |
| Default    | T                 |
| Deprecated | No                |

Output lots of info.

#### Schema

T |X| L

### 18.22 siesta.MaxSCFIterations

|         |                         |
|---------|-------------------------|
| Keyword | siesta.MaxSCFIterations |
| Kind    | scalar                  |
| Type    | Integer                 |

|            |     |
|------------|-----|
| Default    | 100 |
| Deprecated | No  |

Max number of SCF iterations allowed.

#### Schema

100 |X| I

### 18.23 siesta.MeshCutoff

|            |                       |
|------------|-----------------------|
| Keyword    | siesta.MeshCutoff     |
| Kind       | vector                |
| Type       | Vector: Float, String |
| Default    | 120.0 Ry              |
| Deprecated | No                    |

Energy that defines the real-space grid. Point spacing scales as  $1/\sqrt{(E_c)}$ .

#### Schema

120.0 Ry |X| F S

### 18.24 siesta.NumberOfAtoms

|            |                       |
|------------|-----------------------|
| Keyword    | siesta.NumberOfAtoms  |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

Indicate number of atoms in unit cell.

#### Schema

|X| I

### 18.25 siesta.NumberOfSpecies

|            |                        |
|------------|------------------------|
| Keyword    | siesta.NumberOfSpecies |
| Kind       | scalar                 |
| Type       | Integer                |
| Default    | Required / No default  |
| Deprecated | No                     |

Indicate number of species in unit cell.

## Schema

|X| I

### 18.26 siesta.PAO.BasisSize

|            |                      |
|------------|----------------------|
| Keyword    | siesta.PAO.BasisSize |
| Kind       | scalar               |
| Type       | String               |
| Default    | DZP                  |
| Deprecated | No                   |

Define basis size. Can be SZ, DZ, SZP, DZP.

## Schema

DZP |X| S

### 18.27 siesta.PAO.BasisType

|            |                      |
|------------|----------------------|
| Keyword    | siesta.PAO.BasisType |
| Kind       | scalar               |
| Type       | String               |
| Default    | split                |
| Deprecated | No                   |

Define basis type split: Split-valence scheme for multiple-zeta. The split is based on different radii. splitgauss: Same as split but using Gaussian functions. nodes: Generalized PAO's. nonodes: The original PAO's are used, multiple-zeta is generated by changing the scale-factors, instead of using the excited orbitals. filteret: Use the filterets as a systematic basis set. The size of the basis set is controlled by the filter cut-off for the orbitals.

## Schema

split |X| S

### 18.28 siesta.SolutionMethod

|            |                       |
|------------|-----------------------|
| Keyword    | siesta.SolutionMethod |
| Kind       | scalar                |
| Type       | String                |
| Default    | diagon                |
| Deprecated | No                    |

Method for solving the eigenvalue problem.

## Schema

diagon |X| S

### 18.29 siesta.SpinPolarized

|            |                       |
|------------|-----------------------|
| Keyword    | siesta.SpinPolarized  |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Include spin in calculation. Value should be yes or no.

## Schema

|X| S

### 18.30 siesta.TD.CoreExcitedAtom

|            |                           |
|------------|---------------------------|
| Keyword    | siesta.TD.CoreExcitedAtom |
| Kind       | scalar                    |
| Type       | Integer                   |
| Default    | Required / No default     |
| Deprecated | No                        |

Set which atom will be the absorbing atom, i.e., which atoms the core-hole will be centered on

## Schema

|X| I

### 18.31 siesta.TD.CorePerturbationCharge

|            |                                  |
|------------|----------------------------------|
| Keyword    | siesta.TD.CorePerturbationCharge |
| Kind       | scalar                           |
| Type       | Float                            |
| Default    | Required / No default            |
| Deprecated | No                               |

Set the core-hole charge.

## Schema

|X| F

### 18.32 siesta.TD.mxPC

|            |                |
|------------|----------------|
| Keyword    | siesta.TD.mxPC |
| Kind       | scalar         |
| Type       | Integer        |
| Default    | 2              |
| Deprecated | No             |

Maximum number of points to use for predictor-corrector algorithm.

#### Schema

2 |X| I

### 18.33 siesta.TD.NumberOfTimeSteps

|            |                             |
|------------|-----------------------------|
| Keyword    | siesta.TD.NumberOfTimeSteps |
| Kind       | scalar                      |
| Type       | Integer                     |
| Default    | Required / No default       |
| Deprecated | No                          |

Specify number of time steps for RT-TDDFT run.

#### Schema

|X| I

### 18.34 siesta.TD.ShapeOfEfield

|            |                         |
|------------|-------------------------|
| Keyword    | siesta.TD.ShapeOfEfield |
| Kind       | scalar                  |
| Type       | String                  |
| Default    | Required / No default   |
| Deprecated | No                      |

Specify the shape of the field for the RT-TDDFT run. Options are: step - Turn off dipole field at time 0. halfstep - Dipole field changes from full strength to 1/2 strength at time 0. delta - Dipole field with time dependence delta(t). const - Turn on dipole field at time 0. sine - Dipole field with damped sine wave time dependence, with optional Gaussian envelope to create quasi-harmonic wave packet. core - turn on core-hole potential at time 0. core2 - GS with full core-hole. Excited states from linear response after.

#### Schema

|X| S

### 18.35 siesta.TD.TimeStep

|            |                       |
|------------|-----------------------|
| Keyword    | siesta.TD.TimeStep    |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

Specify the time step for RT-TDDFT run.

#### Schema

|X| F

### 18.36 siesta.UseSaveData

|            |                    |
|------------|--------------------|
| Keyword    | siesta.UseSaveData |
| Kind       | scalar             |
| Type       | Logical            |
| Default    | F                  |
| Deprecated | No                 |

Use saved data to start siesta calculation

#### Schema

F |X| L

### 18.37 siesta.XC.authors

|            |                   |
|------------|-------------------|
| Keyword    | siesta.XC.authors |
| Kind       | scalar            |
| Type       | String            |
| Default    | PBE               |
| Deprecated | No                |

Set authors of exchange correlation functional.

#### Schema

PBE |X| S

### 18.38 siesta.XC.functional

|            |                      |
|------------|----------------------|
| Keyword    | siesta.XC.functional |
| Kind       | scalar               |
| Type       | String               |
| Default    | GGA                  |
| Deprecated | No                   |

Set flavor of exchange-correlation functional, i.e., LDA or GGA.

#### Schema

GGA |X| S

## 19 cif2cell

### 19.1 cif2cell.atomicunits

|            |                       |
|------------|-----------------------|
| Keyword    | cif2cell.atomicunits  |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

print coordinates in bohr rather than angstroms

#### Schema

|X| L

### 19.2 cif2cell.cartesian

|            |                       |
|------------|-----------------------|
| Keyword    | cif2cell.cartesian    |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

make output in cartesian form

#### Schema

|X| L

### 19.3 cif2cell.cif\_input

|            |                       |
|------------|-----------------------|
| Keyword    | cif2cell.cif_input    |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Crystallographic information file to use.

#### Schema

|X| S

### 19.4 cif2cell.outfile

|            |                       |
|------------|-----------------------|
| Keyword    | cif2cell.outfile      |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Write output to outfile

#### Schema

|X| S

### 19.5 cif2cell.program

|            |                       |
|------------|-----------------------|
| Keyword    | cif2cell.program      |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

#### Schema

% Program for cif2cell to write input for.  
|X| S

### 19.6 cif2cell.supercell

|         |                    |
|---------|--------------------|
| Keyword | cif2cell.supercell |
| Kind    | vector             |

|            |                                   |
|------------|-----------------------------------|
| Type       | Vector: Integer, Integer, Integer |
| Default    | Required / No default             |
| Deprecated | No                                |

#### Schema

```
|X| I I I
```

## 20 Loop Workflows

### 20.1 loop\_labels

|            |                                       |
|------------|---------------------------------------|
| Keyword    | loop_labels                           |
| Kind       | repeating-record                      |
| Type       | Repeating record with columns: String |
| Default    | Required / No default                 |
| Deprecated | No                                    |

define the labels that will be used for the loop directories. There should be one label for each value of the loop parameter.

#### Schema

```
|X| S
|X| ...
```

### 20.2 loop\_parameter

|            |                                       |
|------------|---------------------------------------|
| Keyword    | loop_parameter                        |
| Kind       | repeating-record                      |
| Type       | Repeating record with columns: String |
| Default    | Required / No default                 |
| Deprecated | No                                    |

Define loop over a set of parameter values for a certain calculation. First line is the keyword to loop over. The rest of the lines are the values associated with each iteration of the loop, with one line per set of values. Note: This only works with single line keywords.

#### Schema

```
|X| S
|X| S ...
|X| ...
```

## 20.3 loop\_target

|            |                       |
|------------|-----------------------|
| Keyword    | loop_target           |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Set the target property to calculate during the loop.

### Schema

|X| S

## 21 Configurational Averaging

### 21.1 cfavg\_choose\_random\_absorbers

|            |                               |
|------------|-------------------------------|
| Keyword    | cfavg_choose_random_absorbers |
| Kind       | scalar                        |
| Type       | Logical                       |
| Default    | False                         |
| Deprecated | No                            |

### Schema

False |X| L

### 21.2 cfavg\_egrid

|            |             |
|------------|-------------|
| Keyword    | cfavg_egrid |
| Kind       | scalar      |
| Type       | String      |
| Default    | "first"     |
| Deprecated | No          |

Choose energy grid type for configurational averaging of spectra. "regular" would be regular in energy, while "first" uses the grid of the first calculation.

### Schema

"first" |X| S

### 21.3 cfavg\_max\_configurations

|            |                          |
|------------|--------------------------|
| Keyword    | cfavg_max_configurations |
| Kind       | scalar                   |
| Type       | Integer                  |
| Default    | 10                       |
| Deprecated | No                       |

Set the number of configurations used in disordered systems.

## Schema

10 |X| I

### 21.4 cfavg\_max\_configurations

|            |                          |
|------------|--------------------------|
| Keyword    | cfavg_max_configurations |
| Kind       | scalar                   |
| Type       | Integer                  |
| Default    | Required / No default    |
| Deprecated | No                       |

## Schema

|X| I

### 21.5 cfavg\_target

|            |                       |
|------------|-----------------------|
| Keyword    | cfavg_target          |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

## Schema

|X| S ...

## 22 Fitting

### 22.1 fit.bond

|            |                                        |
|------------|----------------------------------------|
| Keyword    | fit.bond                               |
| Kind       | repeating-record                       |
| Type       | Repeating record with columns: Integer |
| Default    | Required / No default                  |
| Deprecated | No                                     |

## Schema

|X| I ...  
|X| ...

### 22.2 fit.datafile

|            |                       |
|------------|-----------------------|
| Keyword    | fit.datafile          |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

## Schema

|X| S

### 22.3 fit.parameters

|            |                                              |
|------------|----------------------------------------------|
| Keyword    | fit.parameters                               |
| Kind       | repeating-record                             |
| Type       | Repeating record with columns: String, Float |
| Default    | Required / No default                        |
| Deprecated | No                                           |

## Schema

|X| S F  
|X| ...

### 22.4 fit.target

|            |                       |
|------------|-----------------------|
| Keyword    | fit.target            |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

## Schema

|X| S

## 23 OCEAN

### 23.1 ocean.abpad

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.abpad           |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

For the calculation of the final state wavefunctions, the convergence of the highest bands can be very slow. The calculation can be sped up by adding some throwaway bands at the top. These bands are not considered when checking for convergence. ocean.abpad adds bands to the calculation. Only used for abinit.

#### Schema

|X| I

### 23.2 ocean.acell

|            |                             |
|------------|-----------------------------|
| Keyword    | ocean.acell                 |
| Kind       | vector                      |
| Type       | Vector: Float, Float, Float |
| Default    | Required / No default       |
| Deprecated | No                          |

Scaling in Bohr for the primitive vectors of the unit cell

#### Schema

|X| F F F

### 23.3 ocean.cnbse.broaden

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.cnbse.broaden   |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

Half? width of broadening in eV for the spectrum. Must be larger than zero. Suggested value is the core-hole broadening.

#### Schema

|X| F

### 23.4 ocean.cnbse.gmres.elist

|            |                         |
|------------|-------------------------|
| Keyword    | ocean.cnbse.gmres.elist |
| Kind       | scalar                  |
| Type       | Float                   |
| Default    | Required / No default   |
| Deprecated | No                      |

List of energies at which to run the GMRES algorithm in eV. Lowest unoccupied state is set at zero without the core-hole.

#### Schema

|X| F ...

### 23.5 ocean.cnbse.gmres.erange

|            |                          |
|------------|--------------------------|
| Keyword    | ocean.cnbse.gmres.erange |
| Kind       | scalar                   |
| Type       | Float                    |
| Default    | Required / No default    |
| Deprecated | No                       |

Use energies at with even spacing specified by emin emax estep

#### Schema

|X| F

### 23.6 ocean.cnbse.gmres.ffff

|            |                        |
|------------|------------------------|
| Keyword    | ocean.cnbse.gmres.ffff |
| Kind       | scalar                 |
| Type       | Float                  |
| Default    | Required / No default  |
| Deprecated | No                     |

Sets convergence criterion for GMRES.

#### Schema

|X| F

### 23.7 ocean.cnbse.gmres.gprc

|            |                        |
|------------|------------------------|
| Keyword    | ocean.cnbse.gmres.gprc |
| Kind       | scalar                 |
| Type       | Float                  |
| Default    | Required / No default  |
| Deprecated | No                     |

Set Lorentzian broadening in Hartree in the GMRES algorithm when pre-conditioning. Default is 0.5, which should be reasonable for K edges.

#### Schema

|X| F

### 23.8 ocean.cnbse.gmres.nloop

|            |                         |
|------------|-------------------------|
| Keyword    | ocean.cnbse.gmres.nloop |
| Kind       | scalar                  |
| Type       | Integer                 |
| Default    | Required / No default   |
| Deprecated | No                      |

Restart GMRES algorithm will restart if the subspace grows to size nloop.

#### Schema

|X| I

### 23.9 ocean.cnbse.mode

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.cnbse.mode      |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Which spectroscopy to calculate: "XAS", "XES" NRIXS and XRS are treated the same as XAS.

#### Schema

|X| S

### 23.10 ocean.cnbse.niter

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.cnbse.niter     |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

Number of Haydock iterations. Default is 100.

#### Schema

|X| I

### 23.11 ocean.cnbse.rad

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.cnbse.rad       |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

One of the screening radius as defined by ocean.screen.shells that will be used in the BSE calculation.

#### Schema

|X| F

### 23.12 ocean.cnbse.solver

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.cnbse.solver    |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Choose between Haydock algorithm (full spectrum, no excitons) or GMRES (single energy, exciton density).

#### Schema

|X| S

### 23.13 ocean.cnbse.spect\_range

|            |                               |
|------------|-------------------------------|
| Keyword    | ocean.cnbse.spect_range       |
| Kind       | vector                        |
| Type       | Vector: Integer, Float, Float |
| Default    | Required / No default         |
| Deprecated | No                            |

Sets energies over which the spectrum will be calculated. nsteps emin emax

#### Schema

|X| I F F

### 23.14 ocean.cnbse.strength

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.cnbse.strength  |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

Sets the strength for the two interaction terms in the BSE Hamiltonian. Default is 1, and for emission 0. This can be changed for analysis.

#### Schema

|X| F

### 23.15 ocean.cnbse.xmesh

|            |                                   |
|------------|-----------------------------------|
| Keyword    | ocean.cnbse.xmesh                 |
| Kind       | vector                            |
| Type       | Vector: Integer, Integer, Integer |
| Default    | Required / No default             |
| Deprecated | No                                |

Wave-functions are converted into the NIST BSE format and condensed onto a grid of size ocean.cnbse.xmesh. Default is guessed in the code.

#### Schema

|X| I I I

### 23.16 ocean.core\_offset

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.core_offset     |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

If true, core-level shift will be calculated, false, no shift, or a number to specify the shift by hand.

#### Schema

|X| S

### 23.17 ocean.degauss

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.degauss         |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

Broadening for the smearing function in Ryd. for metallic occupations in the DFT calculations.

#### Schema

|X| F

### 23.18 ocean.dft

|            |           |
|------------|-----------|
| Keyword    | ocean.dft |
| Kind       | scalar    |
| Type       | String    |
| Default    | abi       |
| Deprecated | No        |

Which DFT code to use: qe - QuantumEspresso, abi - ABINIT

#### Schema

abi |X| S

### 23.19 ocean.dft\_energy\_range

|            |                        |
|------------|------------------------|
| Keyword    | ocean.dft_energy_range |
| Kind       | scalar                 |
| Type       | Float                  |
| Default    | Required / No default  |
| Deprecated | No                     |

Instead of setting the number of bands (ocean.nbands) the user may request an energy range in eV for the final state wave-functions. This is an estimate using the volume of the unit cell and may be unreliable. Default is 25 eV.

#### Schema

|X| F

### 23.20 ocean.diemac

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.diemac          |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

Macroscopic dielectric matrix.

#### Schema

|X| F

### 23.21 ocean.ecut

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.ecut            |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

Plane wave basis truncation energy in Rydberg

#### Schema

|X| F

### 23.22 ocean.edges

|            |                                                          |
|------------|----------------------------------------------------------|
| Keyword    | ocean.edges                                              |
| Kind       | repeating-record                                         |
| Type       | Repeating record with columns: Integer, Integer, Integer |
| Default    | Required / No default                                    |
| Deprecated | No                                                       |

Specify the atoms and edges to calculate. Each edge entry consists of 3 integers. The first, if greater than zero, is the index of the atom for which the edge should be calculated. The second and third numbers are the principle quantum number and angular momentum. '1 2 1' denotes the L edges of the first atom. If the first number is negative, then it is interpreted as -Z, where Z is the atomic number, and in that case edges of all atoms with that atomic number will be calculated. '-22 2 1' will run the L edges of every titanium atom in the cell.

#### Schema

```
|X| I I I
|X| ...
```

### 23.23 ocean.fband

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.fband           |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

Determines the number of occupied bands to be included in the SCF calculation of the density. For insulators the default (fband = 0.125) is fine, but for metals the highest band in the density calculation should have no occupation weight. Using fband, the number of bands is determined by  $n = n_{\text{atom}} * \text{fband}$ .

#### Schema

```
|X| F
```

### 23.24 ocean.k0

|            |                             |
|------------|-----------------------------|
| Keyword    | ocean.k0                    |
| Kind       | vector                      |
| Type       | Vector: Float, Float, Float |
| Default    | Required / No default       |
| Deprecated | No                          |

DFT states are calculated using a shifted k-point grid. The first k-point is given by:  $1/(N_k(1) * k0(1))$ ,  $1/(N_k(2) * k0(2))$ ,  $1/(N_k(3) * k0(3))$  where  $N_k$  is given by ocean.nkpt. The default shift is  $k0 = 1/8, 2/8, 3/8$ .

## Schema

|X| F F F

### 23.25 ocean.ldau

|            |                                        |
|------------|----------------------------------------|
| Keyword    | ocean.ldau                             |
| Kind       | repeating-record                       |
| Type       | Repeating record with columns: Logical |
| Default    | Required / No default                  |
| Deprecated | No                                     |

Controls the U parameters for the DFT run. Only works with QuantumEspresso currently. Not sure what the parameters are, but I believe they are logical flag to run/not run LDA+U, U for each atom that has spin associated with smag.

## Schema

|X| L  
|X| F  
|X| ...

### 23.26 ocean.metal

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.metal           |
| Kind       | scalar                |
| Type       | Logical               |
| Default    | Required / No default |
| Deprecated | No                    |

If True, the code will complain unlessoccopt is 3. If False, the codes expectsoccopt = 1.

## Schema

|X| L

### 23.27 ocean.natom

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.natom           |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

Number of atoms in the unit cell

## Schema

|X| I

### 23.28 ocean.nbands

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.nbands          |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

Number of bands for the final state calculations. This includes valence and conduction bands.

## Schema

|X| I

### 23.29 ocean.ngkpt

|            |                                   |
|------------|-----------------------------------|
| Keyword    | ocean.ngkpt                       |
| Kind       | vector                            |
| Type       | Vector: Integer, Integer, Integer |
| Default    | Required / No default             |
| Deprecated | No                                |

Number of k-points in each direction used to sample the cell for the ground-state calculation of the density.

## Schema

|X| I I I

### 23.30 ocean.nkpt

|            |                                   |
|------------|-----------------------------------|
| Keyword    | ocean.nkpt                        |
| Kind       | vector                            |
| Type       | Vector: Integer, Integer, Integer |
| Default    | Required / No default             |
| Deprecated | No                                |

Number of k-points in each direction used to sample the cell for the calculation of the final states.

## Schema

|X| I I I

### 23.31 ocean.nspin

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.nspin           |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

Choose paramagnetic (nspin = 1), or spin dependent (nspin = 2).

#### Schema

|X| I

### 23.32 ocean.nstep

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.nstep           |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

Maximum number of iterations for the DFT SCF calculations. Default is 50

#### Schema

|X| I

### 23.33 ocean.ntypat

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.ntypat          |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

Number of different types (usually species) of atom in the unit cell.

#### Schema

|X| I

### 23.34 ocean.occopt

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.occopt          |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

Controls determination of occupation. Most important values are: 1 - States are all doubly degenerate and either occupied or empty depending on band. 3 - States are all doubly degenerate but can have fractional occupations depending on Fermi function. Suitable for metals.

#### Schema

|X| I

### 23.35 ocean.opf.fill

|            |                         |
|------------|-------------------------|
| Keyword    | ocean.opf.fill          |
| Kind       | vector                  |
| Type       | Vector: Integer, String |
| Default    | Required / No default   |
| Deprecated | No                      |

atomic number followed by the name of the .fill file to use for this calculation.

#### Schema

|X| I S

### 23.36 ocean.opf.hfkgrid

|            |                          |
|------------|--------------------------|
| Keyword    | ocean.opf.hfkgrid        |
| Kind       | vector                   |
| Type       | Vector: Integer, Integer |
| Default    | Required / No default    |
| Deprecated | No                       |

The first parameter sets the grid for hfk.x, and should be left at 2000. The second determines the maximum number of projectors per angular momentum channel. The default of 20 will work for many calculations.

#### Schema

|X| I I

### 23.37 ocean.opf.opts

|            |                         |
|------------|-------------------------|
| Keyword    | ocean.opf.opts          |
| Kind       | vector                  |
| Type       | Vector: Integer, String |
| Default    | Required / No default   |
| Deprecated | No                      |

atomic number followed by the name of the .opts file to use for this calculation.

#### Schema

|X| I S

### 23.38 ocean.opf.program

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.opf.program     |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Use oncvpsp UPS pseudopotentials with qe

#### Schema

|X| S

### 23.39 ocean.para\_prefix

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.para_prefix     |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Set the command to use for parallel runs, i.e., mpirun, mpiexec etc, along with any flags to use.

#### Schema

|X| S ...

### 23.40 ocean.photon.energy

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.photon.energy   |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

Set energy of x-rays at edge

#### Schema

|X| F

### 23.41 ocean.photon.operator

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.photon.operator |
| Kind       | scalar                |
| Type       | String                |
| Default    | dipole                |
| Deprecated | No                    |

Select the operator to be used to calculate the (dipole or quad)

#### Schema

dipole |X| S

### 23.42 ocean.photon.polarization

|            |                             |
|------------|-----------------------------|
| Keyword    | ocean.photon.polarization   |
| Kind       | vector                      |
| Type       | Vector: Float, Float, Float |
| Default    | Required / No default       |
| Deprecated | No                          |

Direction of polarization for XANES or XES calculations

#### Schema

|X| F F F

### 23.43 ocean.photon.qhat

|            |                             |
|------------|-----------------------------|
| Keyword    | ocean.photon.qhat           |
| Kind       | vector                      |
| Type       | Vector: Float, Float, Float |
| Default    | Required / No default       |
| Deprecated | No                          |

Direction of q-vector for quadrupole or NRIXS calculations

#### Schema

|X| F F F

### 23.44 ocean.photon\_q

|            |                             |
|------------|-----------------------------|
| Keyword    | ocean.photon_q              |
| Kind       | vector                      |
| Type       | Vector: Float, Float, Float |
| Default    | Required / No default       |
| Deprecated | No                          |

For UV/VIS and RIXS calculations, the valence and conduction band states must be offset from each other (by the momentum that is absorbed or transferred). This sets that momentum offset in units of the reciprocal lattice vectors of the system. Default is 0 0 0.

#### Schema

|X| F F F

### 23.45 ocean.pp\_list

|            |                                       |
|------------|---------------------------------------|
| Keyword    | ocean.pp_list                         |
| Kind       | repeating-record                      |
| Type       | Repeating record with columns: String |
| Default    | Required / No default                 |
| Deprecated | No                                    |

Names of the pseudopotential files to be used listed in the same order as ocean.znucl

#### Schema

|X| S  
|X| ...

### 23.46 ocean.rprim

|            |                               |
|------------|-------------------------------|
| Keyword    | ocean.rprim                   |
| Kind       | matrix                        |
| Type       | 3 rows of Float, Float, Float |
| Default    | Required / No default         |
| Deprecated | No                            |

Primitive vectors of unit cell

#### Schema

```
|X| F F F
|X| F F F
|X| F F F
```

### 23.47 ocean.scfac

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.scfac           |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

The is the Slater integral factor to scale the slater integrals. For 3d transition metals 0.8 is a reasonable value, while 0.6 is better for f-electron atoms.

#### Schema

```
|X| F
```

### 23.48 ocean.scratch

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.scratch         |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Where to write scratch files

#### Schema

```
|X| S
```

### 23.49 ocean.screen.nbands

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.screen.nbands   |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

Number of bands to use for the screening calculations. This requires a large number of bands. Should be about 100eV above the Fermi level, but convergence should be checked.

#### Schema

|X| I

### 23.50 ocean.screen.nkpt

|            |                                   |
|------------|-----------------------------------|
| Keyword    | ocean.screen.nkpt                 |
| Kind       | vector                            |
| Type       | Vector: Integer, Integer, Integer |
| Default    | Required / No default             |
| Deprecated | No                                |

Number of k-point to be used for the screening calculations. Default is 2 2 2, and is sufficient for a wide variety of systems, although very small unit cells might require more, and large unit cells may only require gamma point.

#### Schema

|X| I I I

### 23.51 ocean.screen.shells

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.screen.shells   |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

The screening calculation is RPA at small radius and a model at large radius. The cross-over between the RPA and model is set by shells in Bohr. Several different radii can be chosen to look at the convergence. Convergence is usually reached at around 3-4 Bohr. The supercell defined by ocean.paw.nkpt must have dimensions larger than the radius specified by ocean.screen.shells or results may be unreliable. The default of 3.5 Bohr is usually OK. Values > 6 Bohr should not be used.

## Schema

|X| F ...

### 23.52 ocean.screen\_energy\_range

|            |                           |
|------------|---------------------------|
| Keyword    | ocean.screen_energy_range |
| Kind       | scalar                    |
| Type       | Float                     |
| Default    | Required / No default     |
| Deprecated | No                        |

Instead of setting number of bands for the screening, the user may request an energy range in eV. This is only an estimate, and may need to be converged.

## Schema

|X| F

### 23.53 ocean.ser\_prefix

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.ser_prefix      |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

prefix for serial parts of the code, for example, cut3d should be run with only 1 processor, i.e.,  
mpirun -n 1

## Schema

|X| S ...

### 23.54 ocean.smag

|            |                                                          |
|------------|----------------------------------------------------------|
| Keyword    | ocean.smag                                               |
| Kind       | repeating-record                                         |
| Type       | Repeating record with columns: Integer, Integer, Integer |
| Default    | Required / No default                                    |
| Deprecated | No                                                       |

Controls initial magnetism for the DFT calculations.

## Schema

|X| I I I

|X| ...

### 23.55 ocean.smag.ixc

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.smag.ixc        |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

Used for abinit runs in smag card. Don't know the meaning at the moment. This should be combined with ocean.smag to write the card to the ocean input.

#### Schema

|X| I

### 23.56 ocean.spin-orbit

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.spin-orbit      |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

If ocean.spin-orbit  $\geq 0$ , the code will not automatically calculate the spin-orbit splitting.

#### Schema

|X| F

### 23.57 ocean.toldfe

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.toldfe          |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

Total energy convergence parameter for the density run. Default is  $10^{-6}$

#### Schema

|X| F

### 23.58 ocean.tolwfr

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.tolwfr          |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

Convergence criterium for the non-scf wave-function calculations. Default is  $10^{-16}$

#### Schema

|X| F

### 23.59 ocean.typat

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.typat           |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

Type of each atom in the unit cell denoted by the integer index corresponding to the ocean.znucl list of atomic numbers. This list must be in the same as the coordinates listed in ocean.xred

#### Schema

|X| I ...

### 23.60 ocean.xred

|            |                                                    |
|------------|----------------------------------------------------|
| Keyword    | ocean.xred                                         |
| Kind       | repeating-record                                   |
| Type       | Repeating record with columns: Float, Float, Float |
| Default    | Required / No default                              |
| Deprecated | No                                                 |

Reduced coordinats of atoms in the unit cell.

#### Schema

|X| F F F  
|X| ...

### 23.61 ocean.znucl

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.znucl           |
| Kind       | scalar                |
| Type       | Integer               |
| Default    | Required / No default |
| Deprecated | No                    |

List of length ntypat with atomic numbers of each type of atom in the unit cell

#### Schema

|X| I ...

### 23.62 ocean.zymb

|            |                       |
|------------|-----------------------|
| Keyword    | ocean.zymb            |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

For use with QE. Each element listed in ocean.znucle can have a symbol associated with it.

#### Schema

|X| S ...

## 24 Miscellaneous

### 24.1 ene\_int

|            |                       |
|------------|-----------------------|
| Keyword    | ene_int               |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

Internal (electronic) energy of the system, in au.

#### Schema

|X| F

## 24.2 mac\_diel\_const

|            |                       |
|------------|-----------------------|
| Keyword    | mac_diel_const        |
| Kind       | scalar                |
| Type       | Float                 |
| Default    | Required / No default |
| Deprecated | No                    |

Approximate value of the macroscopic dielectric constant of the system used in some methods to accelerate convergence of the SCF cycle.

### Schema

|X| F

## 24.3 magmom\_by\_label

|            |                                              |
|------------|----------------------------------------------|
| Keyword    | magmom_by_label                              |
| Kind       | repeating-record                             |
| Type       | Repeating record with columns: String, Float |
| Default    | Required / No default                        |
| Deprecated | No                                           |

set spin for elements of a cif file by the label

### Schema

|X| S F  
|X| ...

## 24.4 mp\_apikey

|            |                       |
|------------|-----------------------|
| Keyword    | mp_apikey             |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Specify your mp API key. This should probably be done with an environment variable instead for security.

### Schema

|X| S

## 24.5 mp\_id

|            |                       |
|------------|-----------------------|
| Keyword    | mp_id                 |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

Single material project id. Can be mp-NNNN or just the number.

### Schema

|X| S

## 24.6 spectralFunction\_file

|            |                       |
|------------|-----------------------|
| Keyword    | spectralFunction_file |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

file to read spectral function from - 2 column file

### Schema

|X| S

## 24.7 xanes\_file

|            |                       |
|------------|-----------------------|
| Keyword    | xanes_file            |
| Kind       | scalar                |
| Type       | String                |
| Default    | Required / No default |
| Deprecated | No                    |

file to read xanes from - 2 column file

### Schema

|X| S